

Password cracker

Module Objectives

- Password - Terminology
- Password Types
- Password Cracker
- How Does a Password Cracker Work?
- Password Cracking Techniques



- Types of Password Attacks
- System Software Password Cracking
- Application Software Password Cracking
- Default Passwords
- Password Cracking Tools



Module Flow



Password
Cracking
Concepts



Types of
Password
Attacks



System
Software
Password
Cracking



Password
Cracking
Tools



Application
Software
Password
Cracking

Password - Terminology

Secret Characters

A password is a secret series of characters that **enables a user to access** a file, computer or a program

A Unique String

It contains a unique string of characters used to **restrict access** to computers and sensitive files



Passwords may contain:

- Only letters **POTHMYDE**
- Only numbers **23698217**
- Only special characters **&*#@!(%)**
- Letters and numbers **meet123**
- Only letters and special characters **bob@&ba**
- Only special characters and numbers **123@\$45**
- Letters, special characters, and numbers **ap1@52**



Password Types

Cleartext Passwords

- A cleartext password is sent over the wire (and wireless) or stored on some media as it is typed without any alteration

Ex: Windows Registry houses automatic logon password
(`HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon`)

- **Cain** and **Ettercap** can be used to sniff cleartext passwords

Obfuscated Passwords

- Passwords stored or communicated after being more or less transformed
- Transformation is reversible, after applying an algorithm the password becomes unreadable and after applying a reverse algorithm it returns to cleartext. This process is called as obfuscation

Hashed Passwords

- Hashed passwords seem similar to obfuscated passwords, but the latter are reversible
- Passwords are transformed using a not-reversible (hashing) algorithm for hashed passwords



Note: Only hashed passwords need cracking, while the other password types can assist in cracking phase

Password Cracker

A password cracker is any program that can **decrypt** passwords or otherwise **disable password protection**

Cracking a **key** means an attempt to recover the **key's value**, and cracking **cipher text** means an attempt to recover the corresponding **plaintext**

Attackers use password **cracking techniques** to gain **unauthorized access** to the vulnerable system

Most of the password **cracking techniques** are successful due to **weak** or **easily guessable** passwords

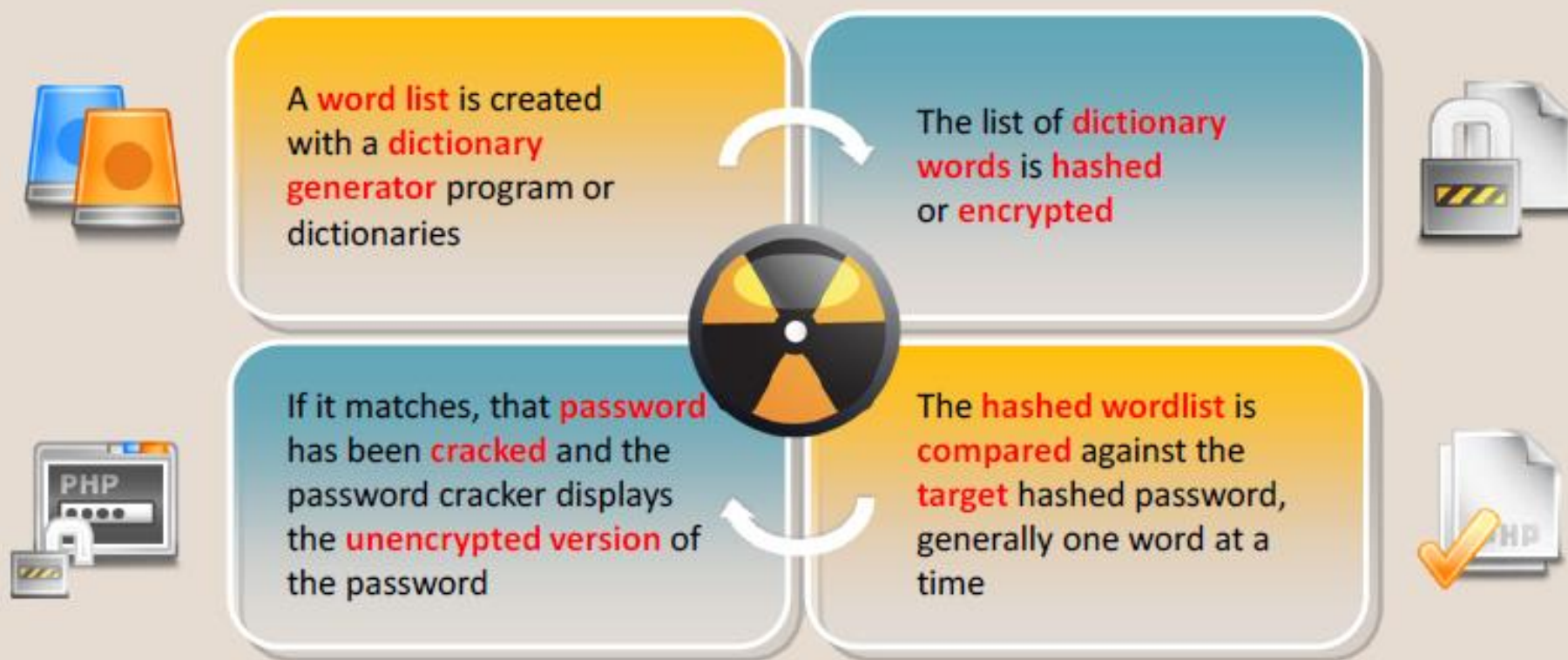


Attacker



Vulnerable System

How Does a Password Cracker Work?



Note: The target hashed password can be obtained by sniffing it from a wired or wireless network or directly from the Security Accounts Manager (SAM) or shadow password files on the hard drive of a system

How Hash Passwords are Stored in Windows SAM



Martin/magician



Password hash using LM/NTLM

Martin:1008:624AAC413795CDC1
4E835F1CD90F4C76:6F585FF8FF6
280B59CCE252FDB500EB8:::

SAM File is located at **c:\windows\system32\config\SAM**

```
Administrator:500:598DDCE2660D3193AAD3B435B51404EE:2D20D252A479F485CDF5E171D93985BF:::  
Guest:501:NO PASSWORD*****:NO PASSWORD*****:::  
HelpAssistant:1000:B991A1DA16C539FE4158440889BE1FFA:2E83DB1AD7FD1DC981F36412863604E9:::  
SUPPORT_388945a0:1002:NO PASSWORD*****:F5C1D381495948F434C42AEE04DE990C:::  
Hackers:1003:37035B1C4AE2B0C5B75E0C8D76954A50:7773C08920232397CAE081704964B786:::  
Admin:1004:NO PASSWORD*****:NO PASSWORD*****:::  
Martin:1005:624AAC413795CDC1AAD3B435B51404EE:C5A237B7E9D8E708D8436B6148A25FA1:::  
John:1006:624AAC413795CDC1FF17365FAF1FFE89:3B1B47E42E0463276E3DED6CEF349F93:::  
Jason:1007:624AAC413795CDC14E835F1CD90F4C76:6F585FF8FF6280B59CCE252FDB500EB8:::  
Smith:1008:624AAC413795CDC14E835F1CD90F4C76:6F585FF8FF6280B59CCE252FDB500EB8:::
```



Username User ID

LM Hash

NTLM Hash

LM hash has been disabled in Windows Vista and Windows 7, LM will be blank in those systems

Module Flow



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Types of
Password
Attacks



System
Software
Password
Cracking



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Password Cracking Techniques



The program tries every combination of characters until the password is cracked

It is the combination of both a brute-force attack and a dictionary attack



Dictionary Attacks

Brute Forcing Attacks

Hybrid Attack

Syllable Attack

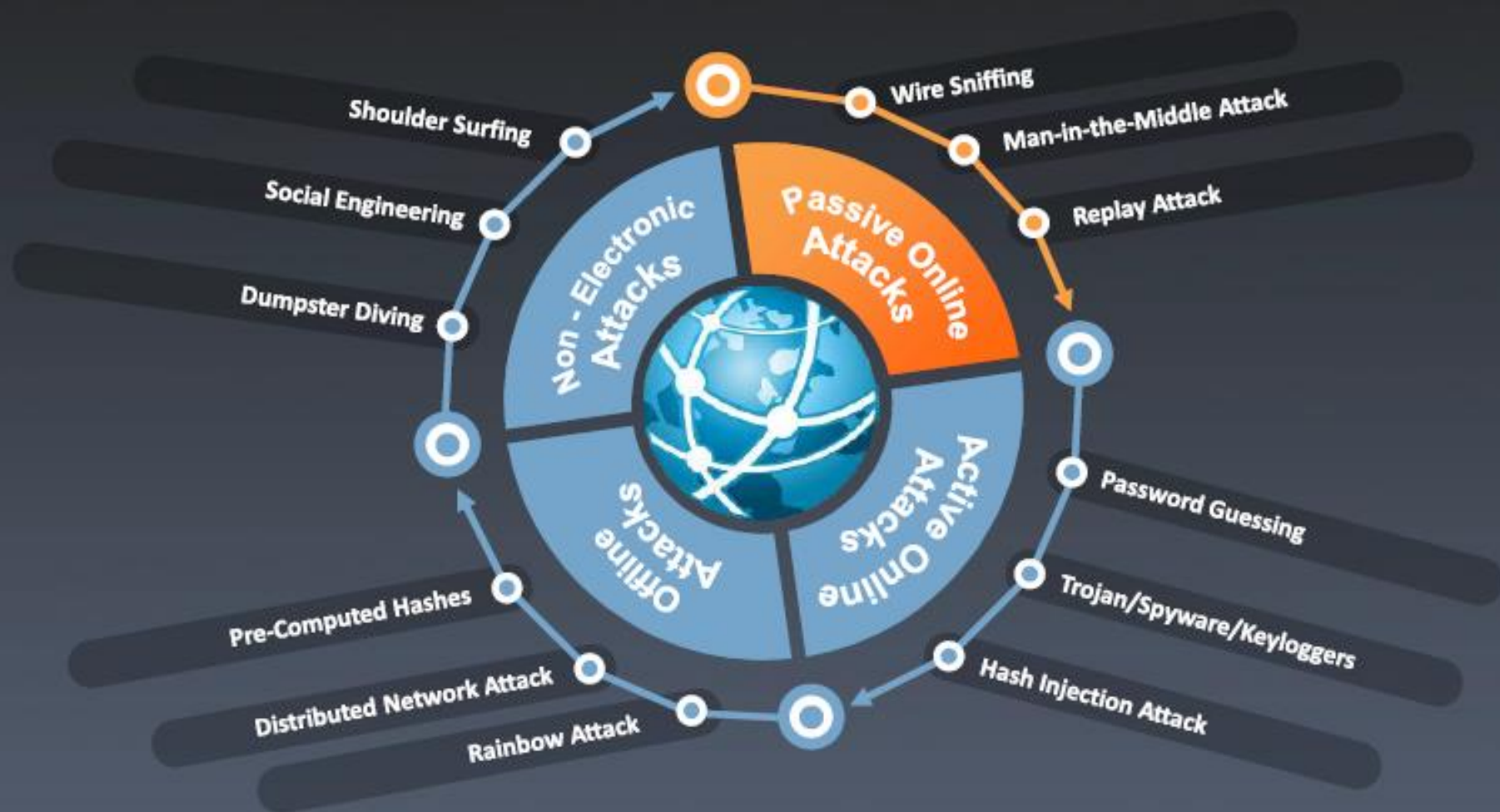
Rule-based Attack

A dictionary file is loaded into the cracking application that runs against user accounts

It works like a dictionary attack, but adds some numbers and symbols to the words from the dictionary and tries to crack the password

This attack is used when the attacker gets some information about the password

Types of Password Attacks



Passive Online Attacks: **Wire Sniffing**

Attackers run packet sniffer tools on the LAN to access and record the raw network traffic



Victim



Attacker



Victim

The captured data may include passwords sent to remote systems during Telnet, FTP, and rlogin sessions, electronic mail sent and received

Hard to Perpetrate

How effective is the attack?

Computationally Complex

Attacker must sniff the network

Tools Available

Password Sniffing

If an attacker is able to eavesdrop on Windows logins, then this approach can spare random guesswork



Sniff credentials off the wire while logging in to a server and then replay them to gain access



Password guessing is a tough task

Passive Online Attack: **Man-in-the-Middle** and **Replay Attack**

MITM

- In man-in-the-middle (MITM) attack, attacker **acquires access to the communication channels** between victim and server to extract the information



Considerations

- Relatively hard to perpetrate
- Must be trusted by one or both sides
- Can sometimes be broken by invalidating traffic

Replay

- In a Replay attack, packets and authentication tokens are captured using a **sniffer**. After the relevant info is extracted, the tokens are placed back on the network to gain access

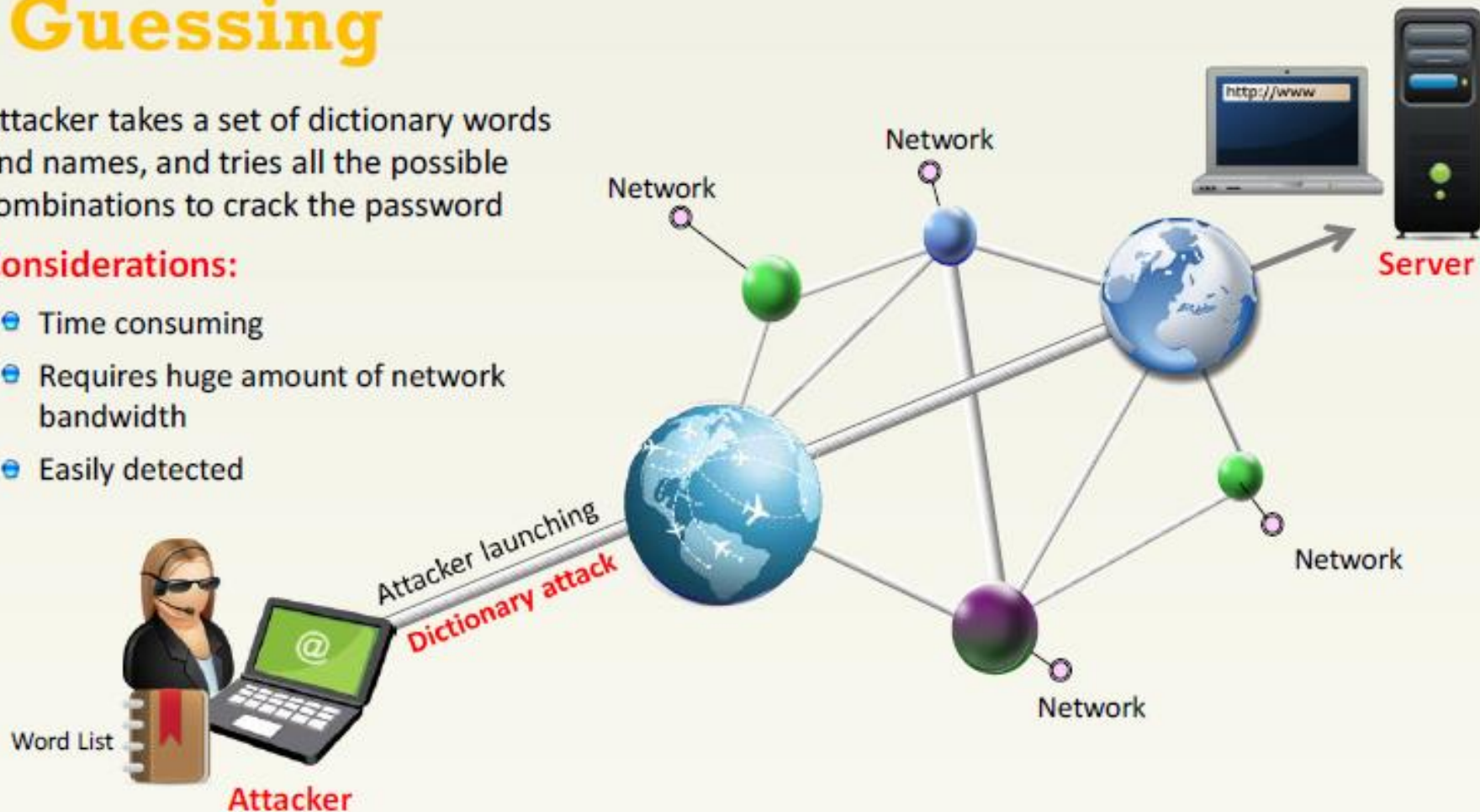


Active Online Attack: Password Guessing

Attacker takes a set of dictionary words and names, and tries all the possible combinations to crack the password

Considerations:

- Time consuming
- Requires huge amount of network bandwidth
- Easily detected



Active Online Attack:

Trojan/Spyware/keylogger

Keylogger

Keylogger is a program that runs in the background and allows a remote attacker to **record every keystroke**



Spyware

Spyware is a type of malware that allows attackers to **secretly** gather personal or organizational information undetected

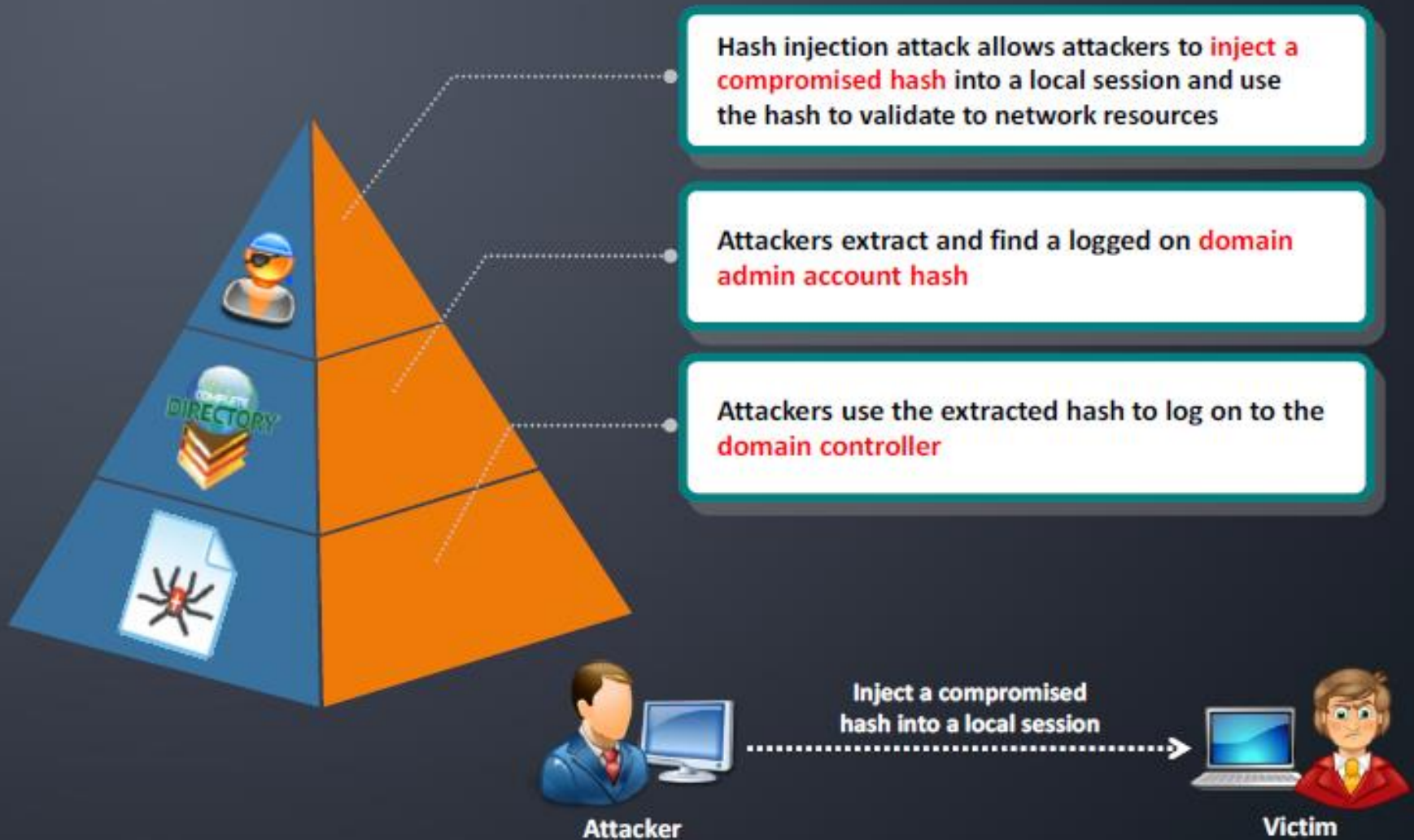


Trojan

With the help of a Trojan, an attacker gets access to the **stored passwords** in the Trojaned computer and would be able to read personal documents, delete files, and display pictures



Active Online Attack: Hash Injection Attack



Rainbow Attacks: Pre-Computed Hash

Rainbow Table

Convert huge word lists i.e. dictionary files and brute-force lists into password hashes using techniques such as rainbow tables

Computed Hashes

Compute the hash for a list of possible passwords and compare it with the precomputed hash table, If a match is found then the password is cracked

Compare the Hashes

It is easy to recover a password by comparing captured password hashes to the precomputed tables

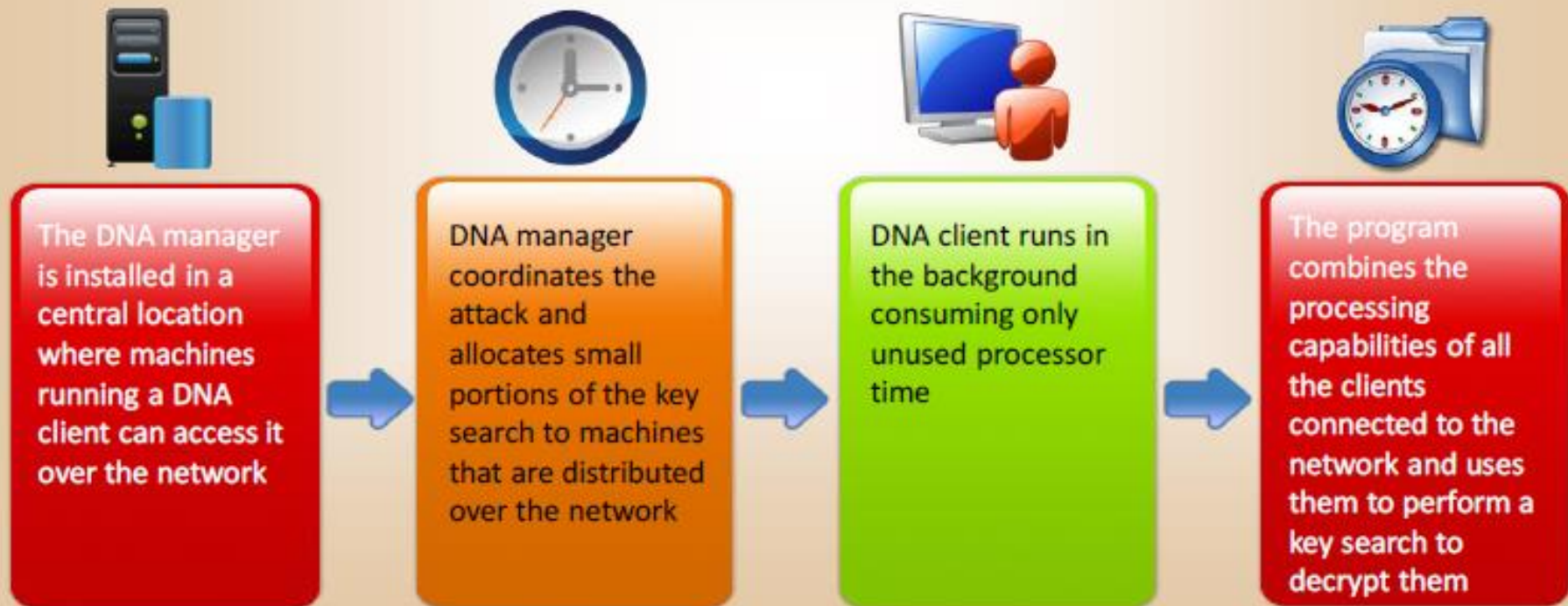
Precomputed Hashes

1qazwed	->	4259cc34599c530b28a6a8f225d668590
hh021da	->	c744b1716cbf8d4dd0ff4ce31a177151
9da8dasf	->	3cd696a8571a843cda453a229d741843
sodifo8sf	->	7ad7d6fa6bb4fd28ab98b3dd33261e8f

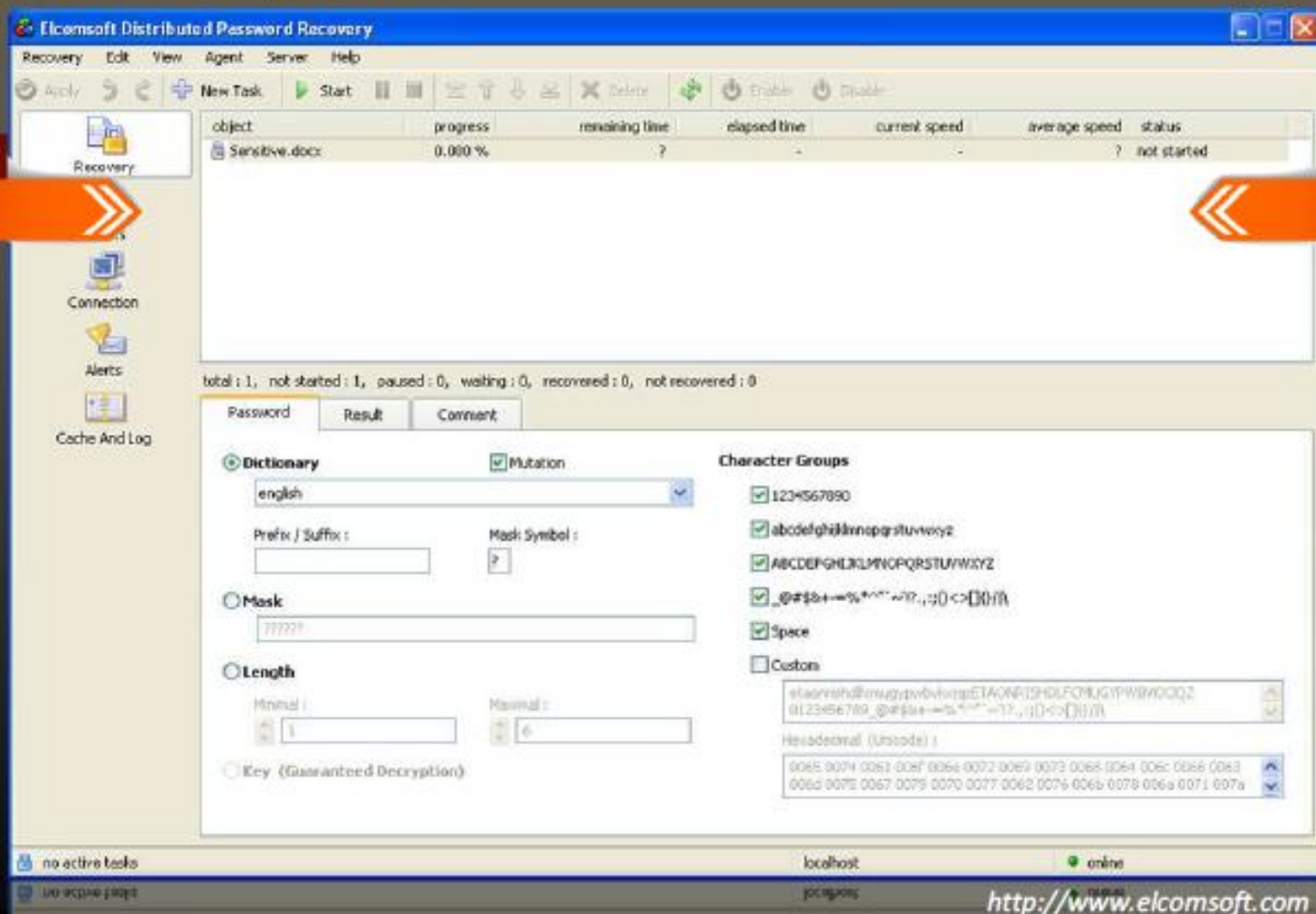


Distributed **Network** Attack

1. Distributed Network Attack (DNA) technique is used for **recovering password-protected files** using the unused processing power of machines across the network to **decrypt passwords**
2. In this attack, a **DNA manager** is installed in a central location where machines running **DNA clients** can access it over the network



Elcomsoft Distributed Password Recovery



Non-Electronic Attacks

Shoulder Surfing



- Looking at either the user's keyboard or screen while he/she is logging in

Dumpster Diving



- Searching for sensitive information in the user's trash bins and printer trash bins, and searching the user's desk for sticky notes

Social Engineering



- Convincing people to reveal the confidential information

Manual Password Cracking (Guessing)



Attacks are **less** frequent

Find a valid user

Create a list of possible passwords

Rank passwords from high probability to low

Key in each password, until correct password is discovered

The failure rate is **high**



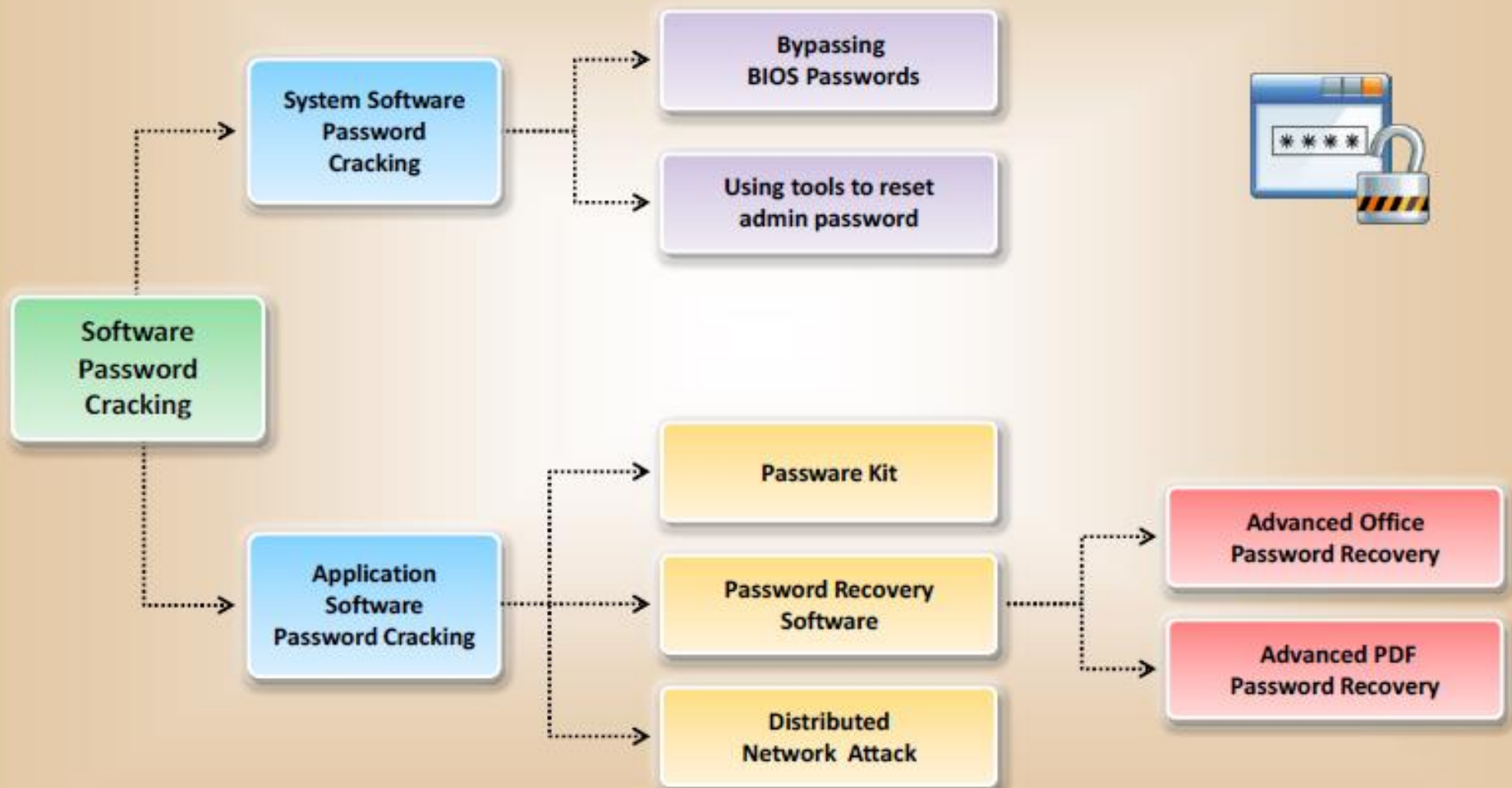
Automatic Password Cracking Algorithm



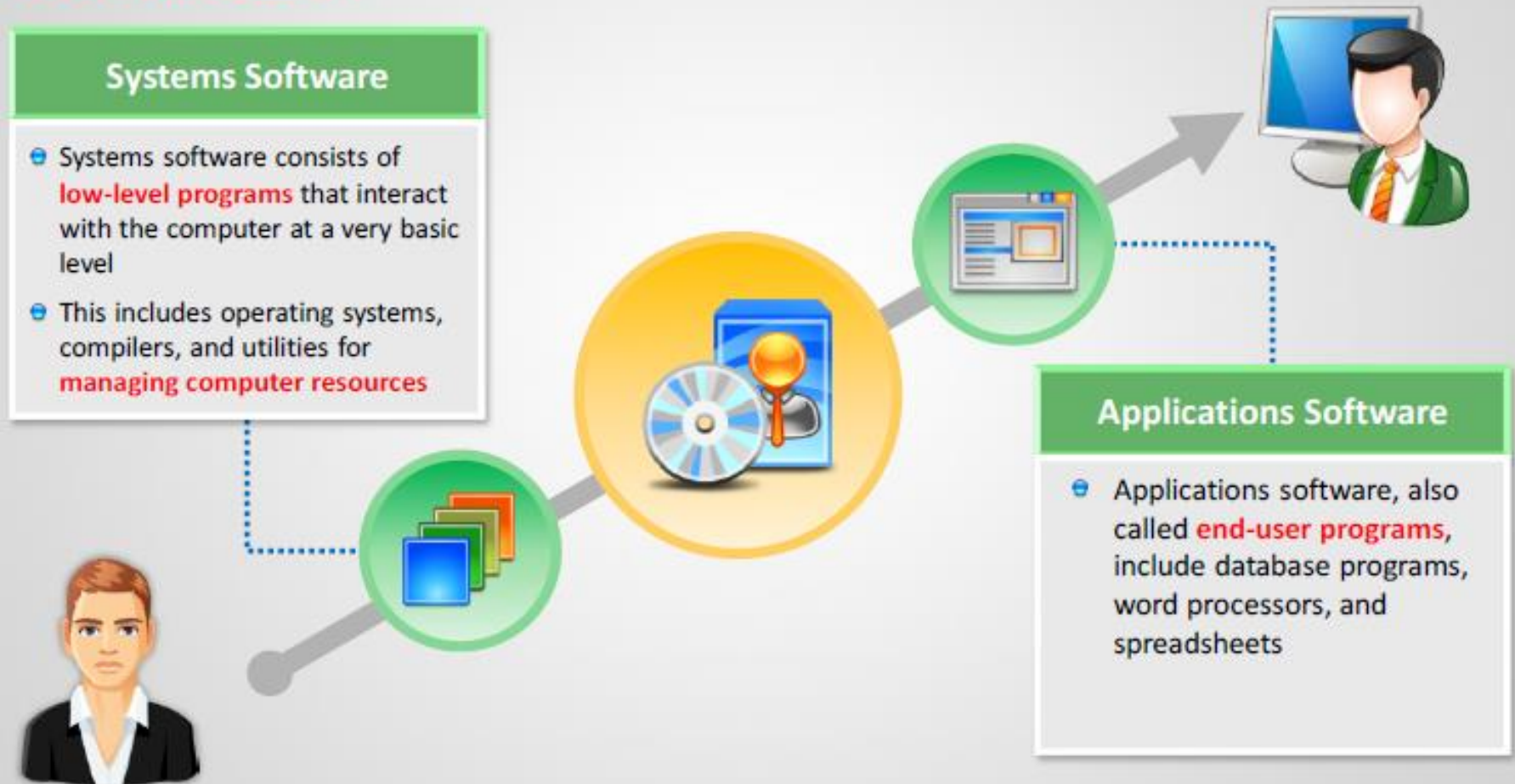
Time Needed to Crack Passwords

Time Needed to Crack Passwords			
Number of characters in password	Total number of characters from which password is selected		
	26 (Lower case letters only -abc)	36 (Lower case letters Plus numbers-abc123)	52 (Upper and lower case letters –AaBbCc)
5	1.98 minutes	10.1 minutes	1.06 hours
6	51.5 minutes	3.74 hours	3.91 months
7	22.3 hours	9.07 days	3.91 months
8	24.2 days	10.7 months	17.0 years
9	1.72 years	32.2 years	8.82 centuries
10	44.8 years	1.16 millennia	45.8 millennia
11	11.6 centuries	41.7 millennia	2,384 millennia
12	30.3 millennia	1,503 millennia	123,946 millennia

Classification of **Cracking Software**



Systems Software vs. Applications Software



Module Flow



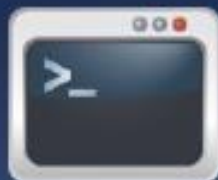
Password
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Types of
Password
Attacks



System
Software
Password
Cracking



Password
Cracking
Tools



Application
Software
Password
Cracking

System Software Password Cracking

System software password cracking is defined as cracking the **operating system** and all other **utilities** that enable a computer to **function**



Passwords are created for the system software to **avoid access** to the system files and other secured **information** used during the booting of the system



Following are the ways to access the system:

- **Bypassing the BIOS password**
- **Using tools to reset admin password**



Bypassing BIOS Passwords

- BIOS (Basic Input Output System) is a **firmware code** run by a system when first **powered on**, which is a type of **boot loader**
- Main **function** of BIOS is to **identify** and **initialize** system component hardware (such as hard disk, floppy drive, and video display card)



Using **Manufacturer's Backdoor** Password to Access the BIOS



- Manufacturer's **provide** a **backup** password that can be used to **access** the **BIOS** setup if the password is **lost**

- The passwords provided by the manufacturers are **case sensitive**. If a particular backdoor password does not work, then various case sensitive **combinations** of the password can be tried.
- The **combinations** may include various combinations including **alphanumeric** characters
- Manufacturers' **documentation** must be **read** before trying the backdoor passwords, because **BIOS** combinations will **lock** the system completely if the password is typed **wrong** three times



A few **manufacturers** and their **backdoor passwords** are listed below:

- VOBIS & IBM – merlin
- Dell – Dell
- Biostar – Biostar
- Compaq – Compaq
- Enox - xo11nE
- EpoX - central
- Freemove - Posterie
- Iwill - iwill
- Jetway - spooml
- Packard Bell - bell9
- QDI - QDI

Using Password Cracking Software

The following software can be used to either crack or reset the BIOS on many chipsets

CmosPwd

Decrypts password stored in CMOS used to access BIOS SETUP



RemPass

Program that can find and show BIOS password, collapse BIOS password, save BIOS settings, and restore BIOS settings



<ftp://ftp.efn.hu>

Note: If your PC is locked with a BIOS administrator password that will not allow access to the floppy drive, these utilities may not work

CmosPwd



CmosPwd decrypts passwords stored in CMOS used to **access BIOS SETUP**

It works with the following BIOSes:

- ACER/IBM BIOS
- AMI BIOS
- AMI WinBIOS 2.5
- Award 4.5x/4.6x/6.0
- Compaq (1992)
- Compaq (New version)
- IBM (PS/2, Activa, Thinkpad)
- Packard Bell
- Phoenix 4 release 6 (User)
- Gateway Solo - Phoenix 4.0 release 6
- Toshiba
- Zenith AMI

```
Command Prompt
C:\cmos>ioperm -i

C:\cmos>cmospwd_win
CmosPwd - BIOS Cracker 5.0, October 2007, Copyright 1996-2007
GRENIER Christophe, grenier@cgsecurity.org
http://www.cgsecurity.org/

Keyboard : US
Acer/IBM           [H] [ < ]
AMI BIOS           []
AMI WinBIOS <12/15/93> []
AMI WinBIOS 2.5     [] [] [] []
AMI ?              [] [ M] [ ] [ M] [ E M]
Award 4.5x6.0       [23012202] [000100] [000300]
Award 4.5x6.0       [30000122] [12003000] [33333033] [0
Award Medallion 6.0 [000100] [000100] [20000000] [00010
Award 6.0           [ [o] [+ 2 S ] [ ] N.P ] [Nq50]
Compaq <1992>        []
Compaq DeskPro      [] []
Compaq               [1 7 ] []
```

<http://www.cgsecurity.org>

Resetting the CMOS using the **Jumpers** or **Solder Beads**

Resetting the CMOS using the Jumpers

- ❌ Motherboard features a set of jumpers or dip switches that clears the **BIOS/CMOS settings** allowing them to be reset
- ❌ Check the computer or motherboard manufacturer's documentation to locate the **jumpers/dip switches**
- ❌ If the document is not available, by default the **jumper position** is across pins 1 and 2
- ❌ Shut down the system and unplug the power cord
- ❌ Move the jumper from its default position so that it is across **pins 2 and 3**; this clears the BIOS/CMOS settings



Resetting the CMOS using the Solder Beads

- ❌ Connecting or jumping specific **solder beads** on the chipset will likely reset the CMOS
- ❌ There are **too many chipsets** to do a breakdown of which points to jump on individual chipsets, and the location of these solder beads can **vary by manufacturer**, so please check the computer and motherboard documentation for details



Removing CMOS Battery



1

Shut down the **system** and **take out** the **plugs** connected to electricity



2

Open the **CPU cabinet** and find the **CMOS battery** (silver circular battery) on the motherboard



3

Remove the **CMOS** battery and keep it out for 2-3 minutes, this **flushes** out the **CMOS memory** that stores BIOS passwords and other configurations

4

Replace the **battery** and **start** the **system** normally



Overloading the **Keyboard Buffer** and Using a **Professional Service**

Overloading the keyboard buffer

- On some older systems, CMOS can be forced to **setup screen on boot** by overloading the keyboard buffer
- This is achieved by hitting the **ESC** key over 100 times repeatedly or by booting with the keyboard or mouse unattached to the systems



Using a professional service

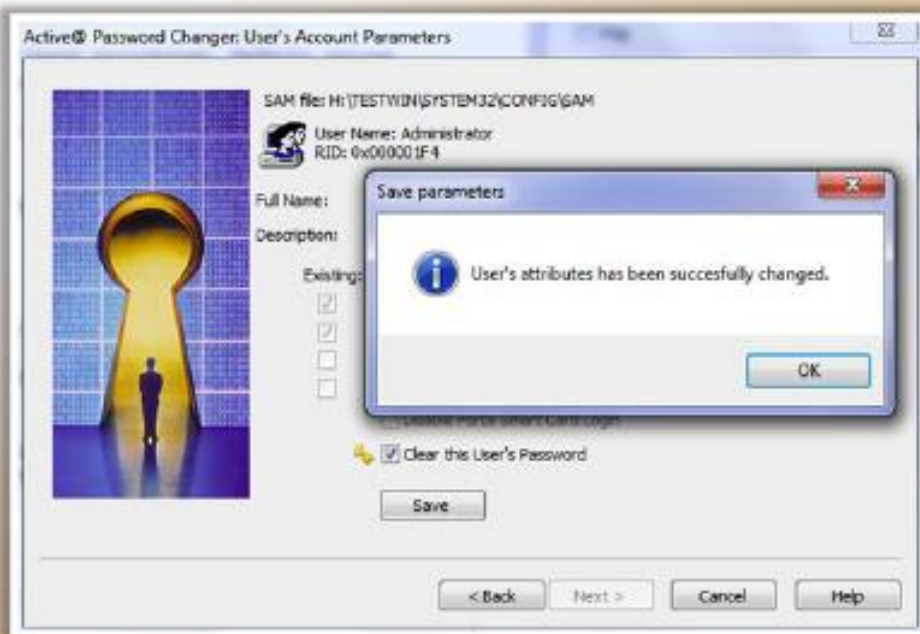
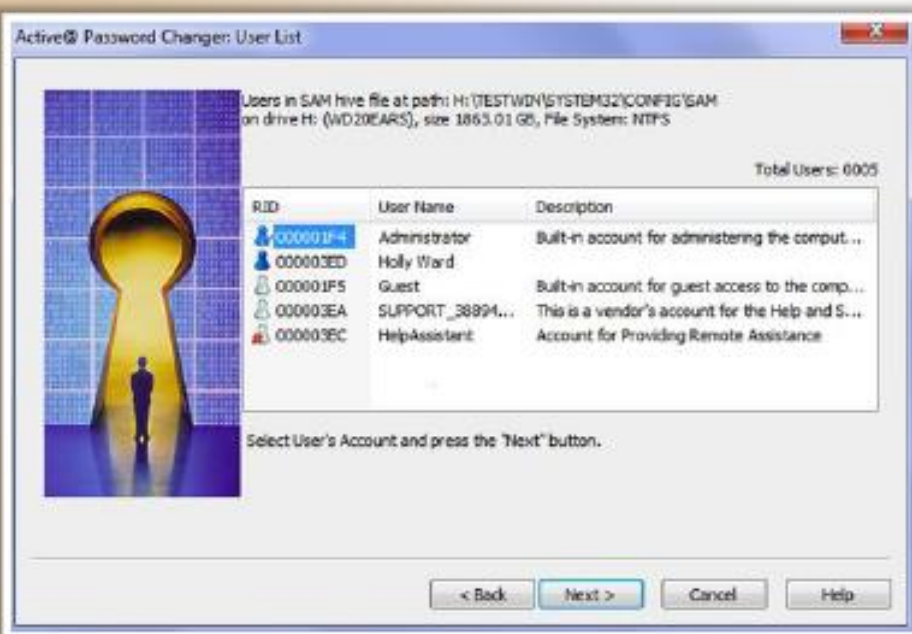
- Professional service can be used if the manufacturer of the laptop or desktop PC would not **reset the password**
- Password Crackers, Inc., offers a variety of services for desktop and laptop computers; all you need to provide is **legitimate proof of ownership**



Tool to Reset Admin Password:

Active@ Password Changer

- Active@ Password Changer is designed for **resetting** local administrator and user passwords on Windows XP / VISTA / 2008 / 2003 / 2000 & Windows 7 systems in case an Administrator's password is **forgotten or lost**
- With Active@ Password Changer you can log in as a particular user with a **blank password**



<http://www.password-changer.com>

Tool to Reset Admin Password: Windows Key



Windows Key **creates** a password reset CD or USB flash drive that works during the **boot process** and instantly resets **Administrator** or **other** account **passwords** and Windows **security** settings that prevent you from logging in

Features

- Resets passwords with a **bootable CD**
- Resets local **Administrator** password
- Resets local **policy** settings
- Resets secure **boot** options
- Displays account **properties**



<http://www.lostpassword.com>

Module Flow



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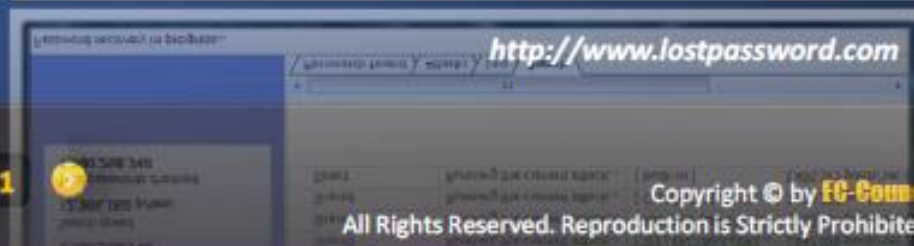
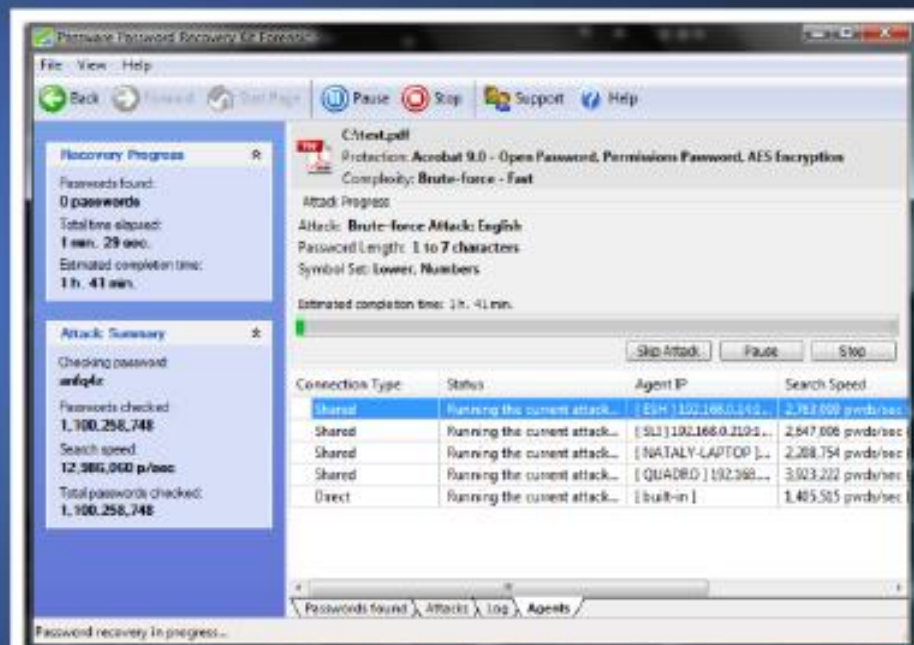
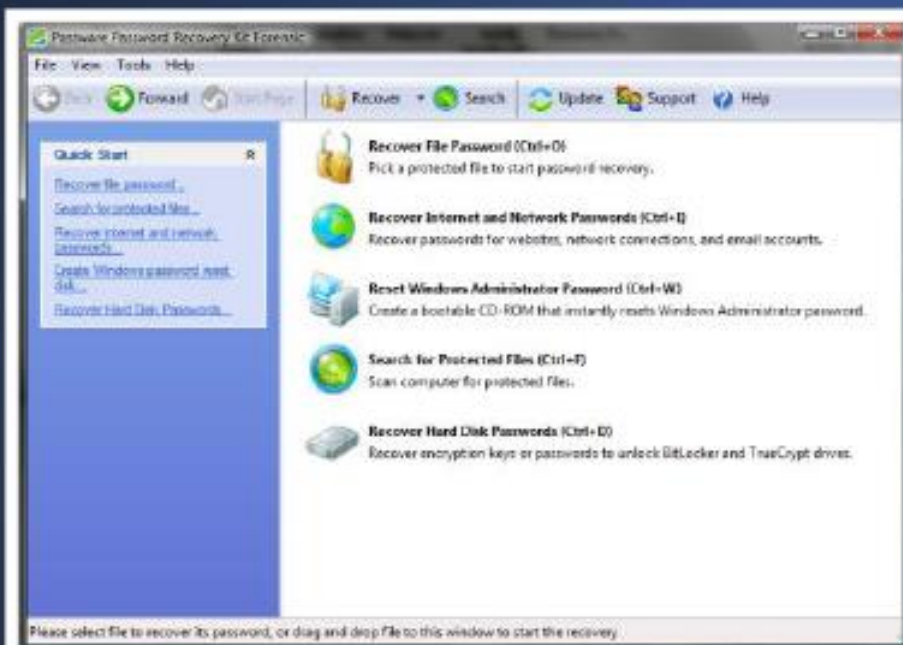
Password
Cracking
Tools



Application
Software
Password
Cracking

Passware Kit Forensic

- This complete electronic evidence discovery solution **reports** all password-protected items on a computer and **gains** access to these items using the fastest **decryption** and password **recovery algorithms**



Accent Keyword Extractor

- Accent Keyword Extractor uses the **dictionary attack** method
- It follows all the links found on the page and also **extracts words** from the linked page



This Program:

- Loads a page from the Internet
- Extracts the **unique** words from the web page, and
- Adds them to the **dictionary**



Distributed Network Attack



Distributed Network Attack, (DNA), is a new technique to **recover the password protected files**

It uses the machines across the network to **decrypt passwords**

The **DNA** server is **installed** in a central location where machines running **DNA client** can access it over the **network**

Features

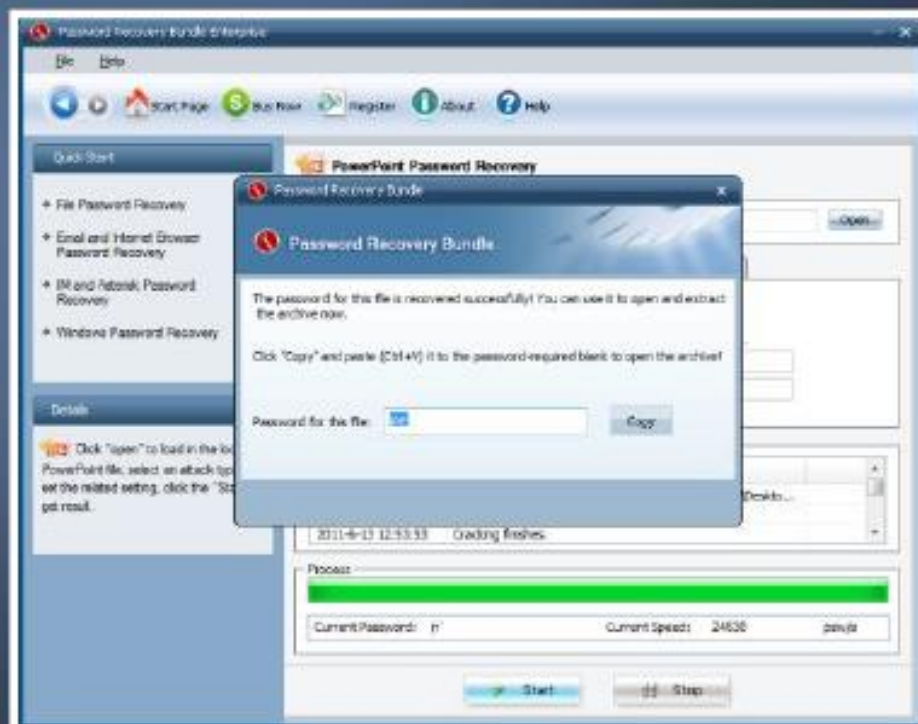
- Stealth client installation **functionality**
- Automatic client **update** when updating the **DNA Server**
- Controls** what clients **work** on certain jobs
- Easy to read **statistics** and **graphs**
- Adds** user dictionaries
- Optimizes** password **attacks** for specific languages
- Customizes** the user's **dictionaries**

<http://accessdata.com>

Password Recovery Bundle



- Password recovery bundle **recovers passwords for Windows** Excel, Word, Access, PowerPoint, Outlook, Outlook Express, PDF, RAR/WinRAR, ZIP/WinZIP, MSN, AOL, Google Talk, Paltalk, Trillian, Miranda, Opera, Firefox and IE Browser, etc.



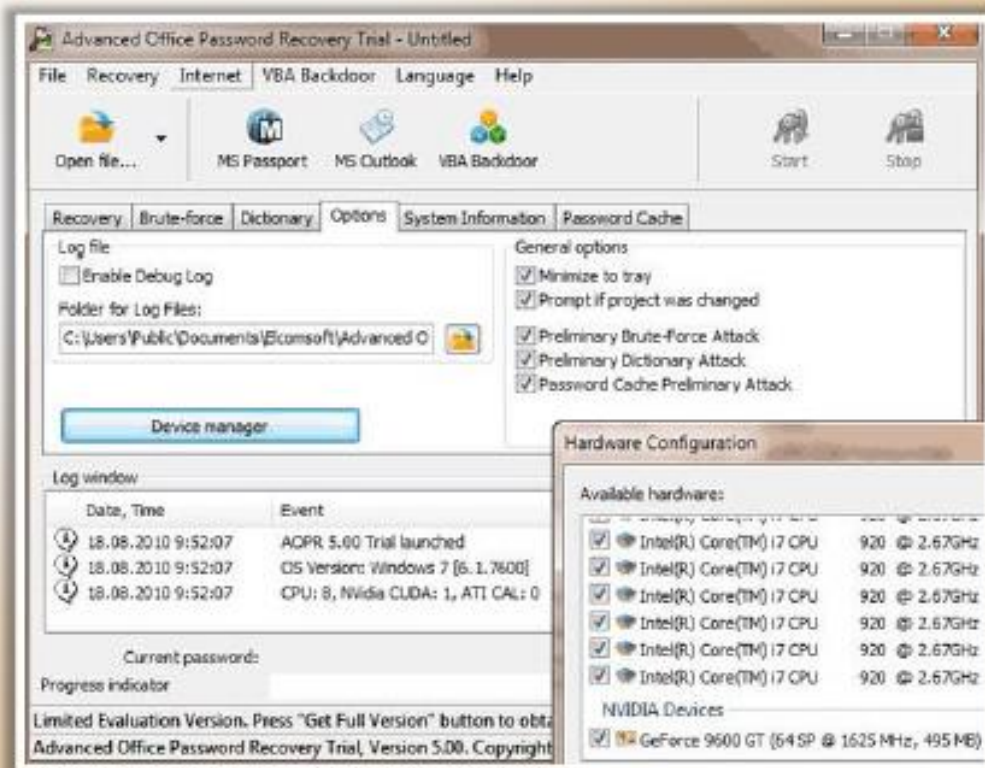
Advanced Office Password Recovery

- Advanced Office Password Recovery **unlocks documents** created with all versions of Microsoft Office from the ancient 2.0 to the modern 2010
- Recovers** passwords for Microsoft Word, Excel, Access, Outlook, Project, Money, PowerPoint, Visio, Publisher and OneNote



Features

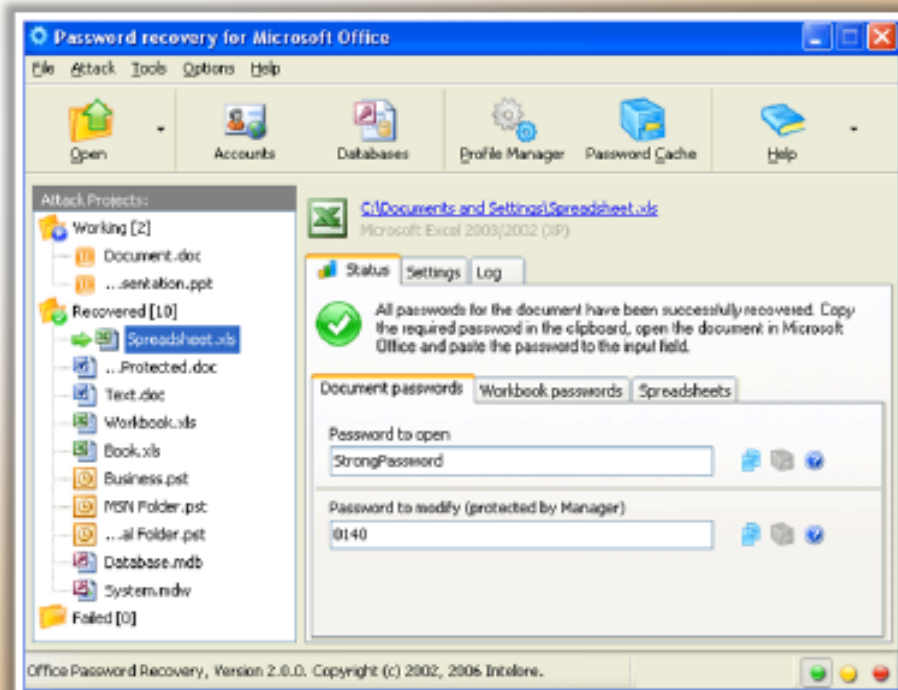
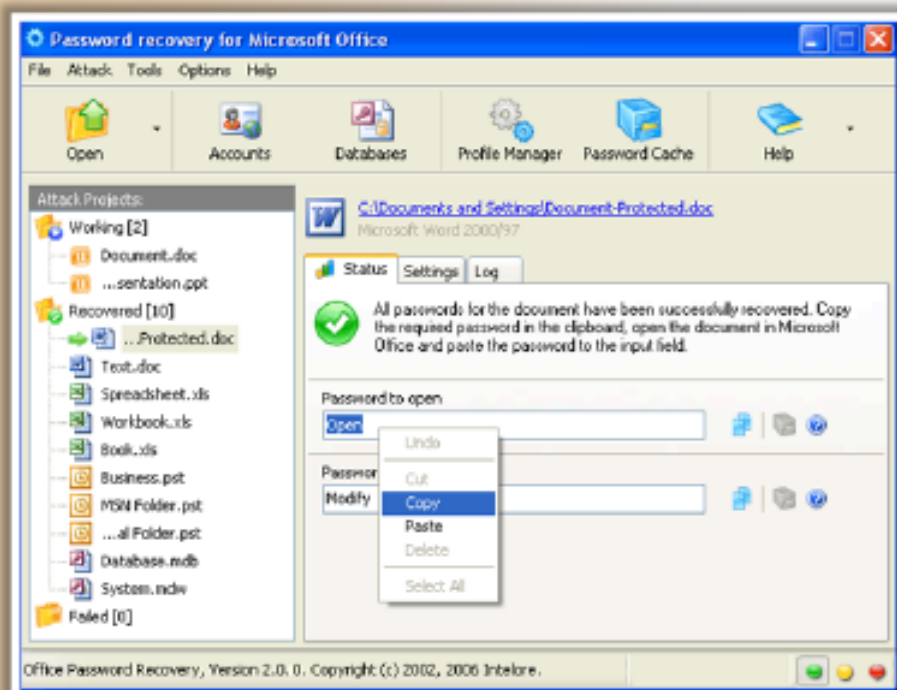
- Instant password **recovery** for multiple **products**
- Exploits** all known backdoors and tricks in the Office family for **instant recovery**
- Instantly **unlocks** documents with **previously** recovered **passwords**
- Dictionary and brute-force **attacks** with **user-defined** masks and advanced templates



<http://www.elcomsoft.com>

Office Password Recovery

- Office Password Recovery **recovers lost or forgotten passwords** for Microsoft Word, Excel, Access, PowerPoint, OneNote and Outlook email accounts, and personal folder files
- It recovers all types of passwords, including instant recovery of passwords to modify, database passwords, workbook passwords, pst passwords, and email account passwords



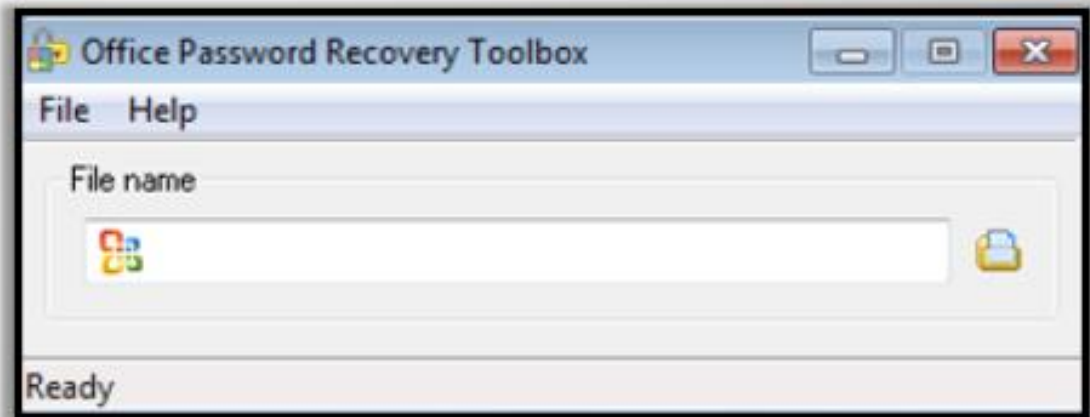
<http://www.passwordrecovery.in>

Office Password Recovery Toolbox

Office Password Recovery Toolbox is a comprehensive solution for **recovering** MS Word, Excel, Outlook, Access, PowerPoint and VBA passwords.

Features

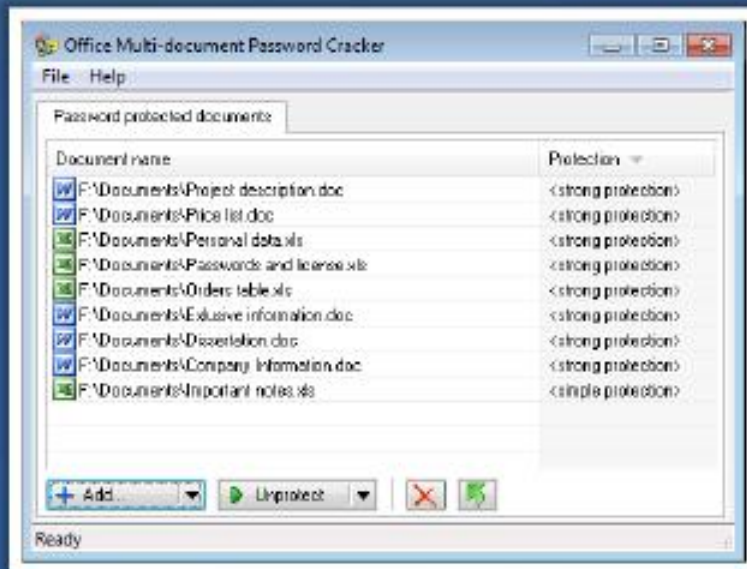
-  **Recovers** or **removes** "open," "write," and "protection" passwords for **Microsoft Word** documents
-  **Recovers** or removes "open," "write," workbook, shared workbook, and worksheet passwords for **MS Excel** documents
-  **Recovers** passwords for the MS Outlook **personal folder** files (*.PST)



<http://www.rixler.com>



Office Multi-document Password Cracker



- Office Multi-document Password Cracker is used to get **instant access to password protected** MS Word and Excel documents
- The program provides a simple solution to **remove any password** from multiple MS Word and MS Excel documents

Features

- Removes the **"open," "write" and "protection" passwords** from MS Word documents
- Removes the **"open," "write," workbook, shared workbook and worksheet** passwords from MS Excel documents
- Allows you to **unprotect multiple documents** with a few mouse clicks
- Protects passwords against **unauthorized program access**

<http://www.rixer.com>

Word Password Recovery Master

- Word Password Recovery Master is used to **crack** password-protected **documents** created in **MS Word** 97/2000/XP/2003/2007/2010
- The program **allows** you to crack "**open**," "**protection**" and "**write**" passwords



Features

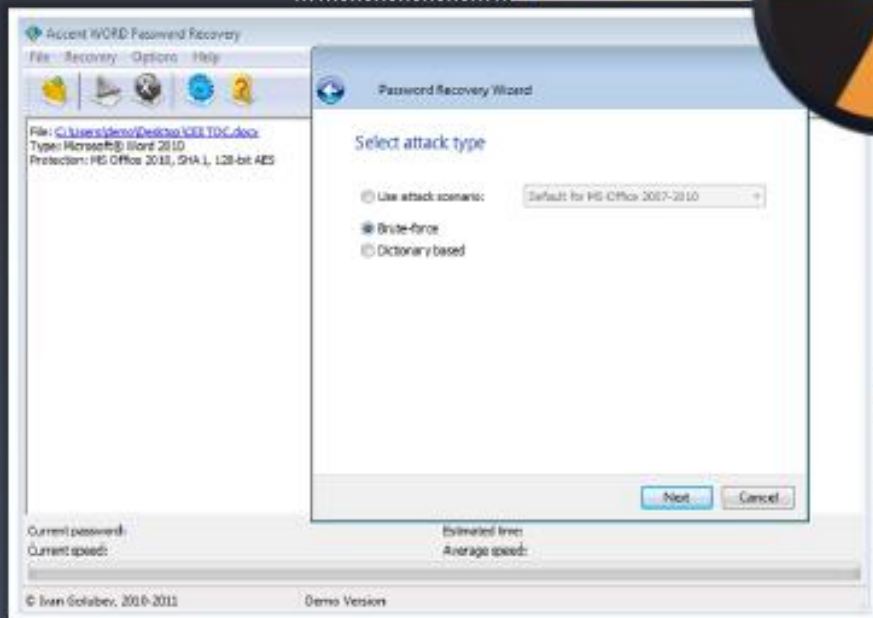
- Removes the "**open**" MS Word password
- Removes the "**protection**" MS Word password
- Recovers the "**write**" MS Word password
- Online** Word password recovery service with guaranteed **privacy**
- Instant recovery** of passwords of any length and complexity
- Support** for **multilingual** passwords



<http://www.rixler.com>

Accent WORD Password Recovery

- Accent WORD Password Recovery **recovers lost Microsoft Word passwords**
- It provides **full support for all versions of Microsoft Word** with lightning-fast recovery of Microsoft Word passwords



Features

- Supports all versions of Microsoft Word (from Word 6/95 to Word 2010)
- Instantly deletes several types of passwords
- Automatically recovers passwords using pre-installed scenarios
- Includes macrolanguage for adding password mutation rules to dictionary attacks
- Reduces password recovery time on all computers with Intel and AMD processors

<http://www.passwordrecoverytools.com>

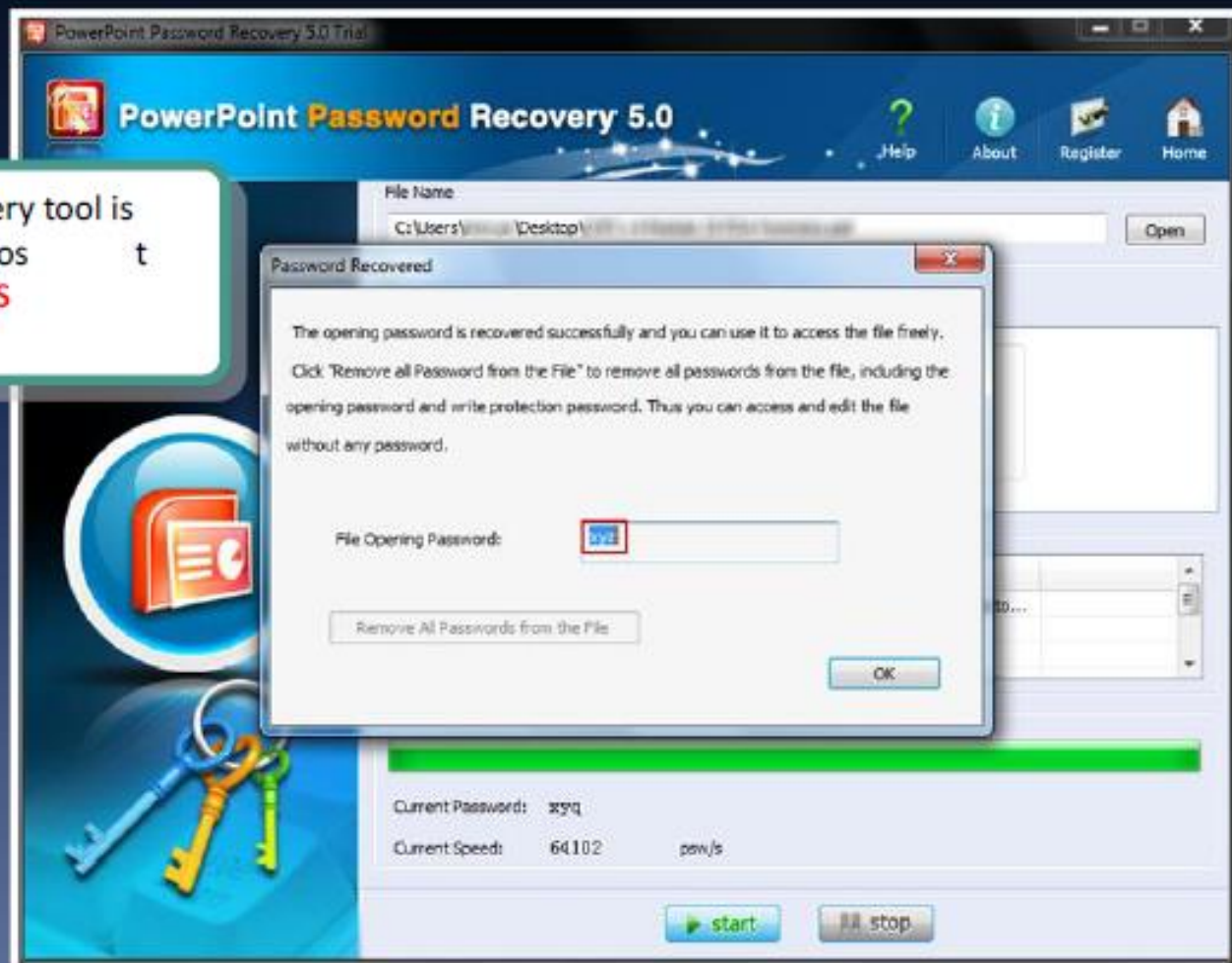
Word Password

- Word Password is able to recover any Word document **regardless of the password length**
- It supports **several protection types**: password to open, password to modify, document protection password, etc.



PowerPoint Password Recovery

PowerPoint Password Recovery tool is designed for recovering the lost or forgotten passwords of **MS PowerPoint presentations**



<http://www.recoverlostpassword.com>

PowerPoint Password

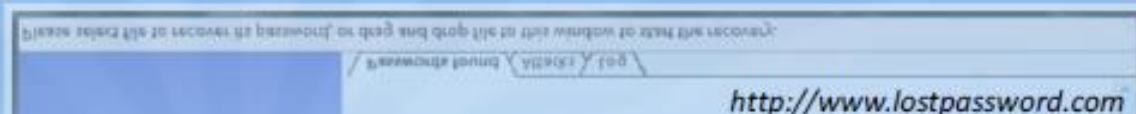
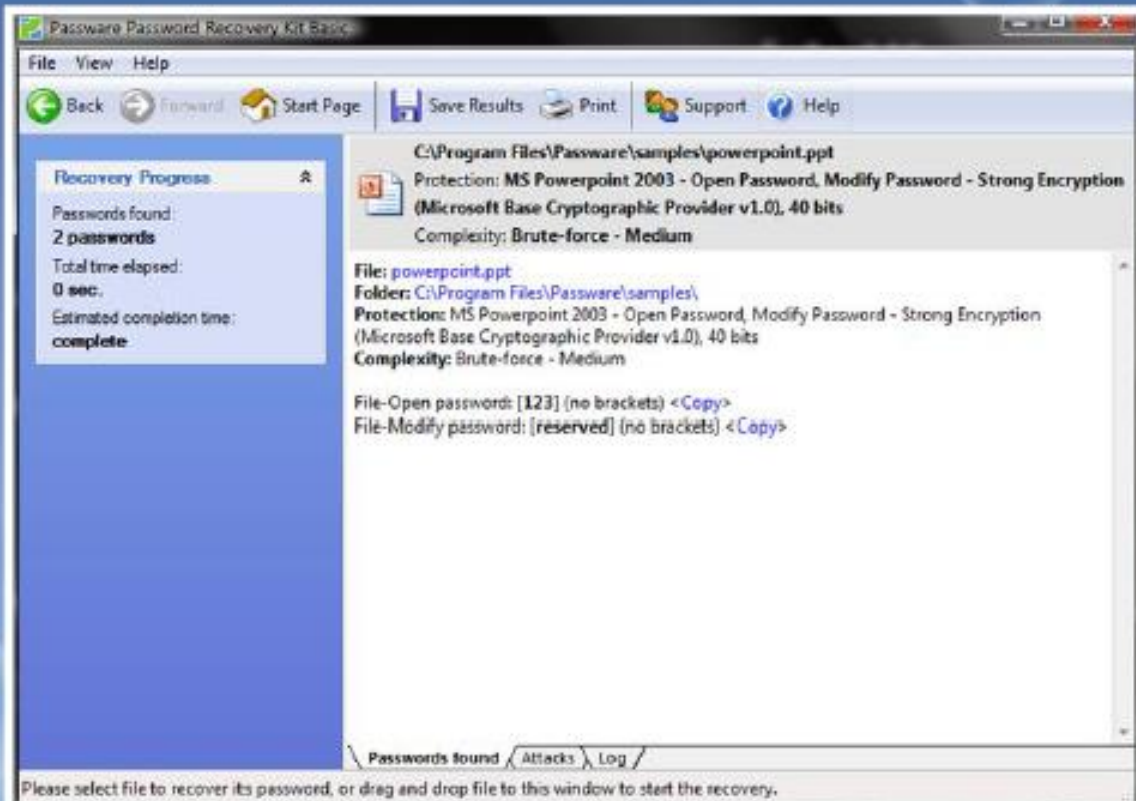
- PowerPoint Password recovers lost or forgotten passwords to PowerPoint (*.ppt) files
- It supports Brute-Force, Dictionary, Hybrid Dictionary, and SmartForce (TM) attacks



<http://lastbit.com>

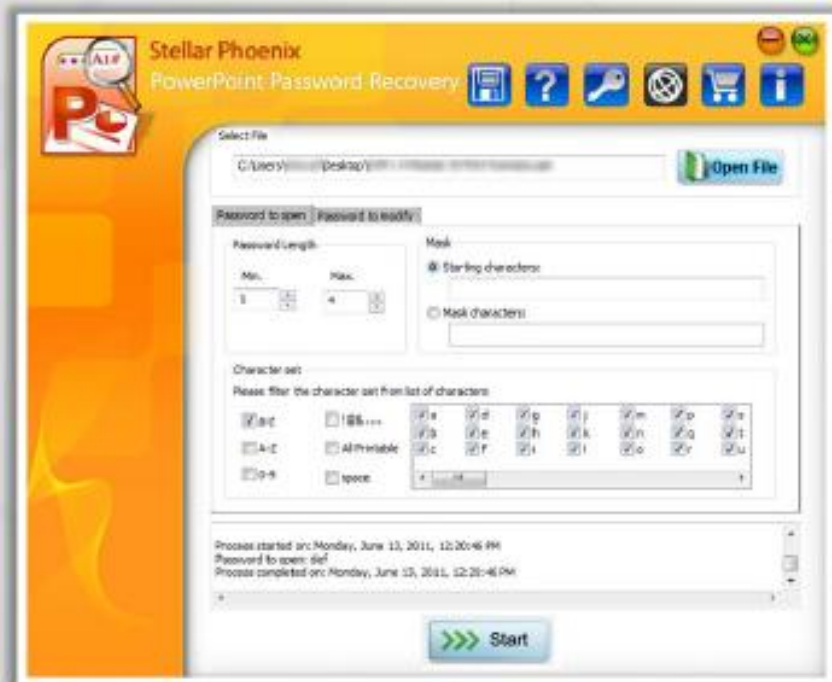
Powerpoint Key

- Powerpoint Key recovers passwords for **MS PowerPoint presentations**
- It recovers **file-open passwords** using a combination of brute-force, Xieve, and dictionary attacks
- It automatically saves the **password search state** and can resume after a stop or a crash
- It supports **non-English characters** in passwords



Stellar Phoenix Powerpoint Password Recovery

- PowerPoint Password recovery software **resets** and **retrieves** PowerPoint passwords using password masking techniques
- It recovers both "**password to open**" and "**password to modify**" PowerPoint presentations from any situation
- It allows you to enter a minimum and maximum number of characters between which it would attempt to **crack the password**



<http://www.stellarinfo.com>

Excel Password Recovery Master



- Excel Password Recovery Master is used to **crack password-protected documents** created in MS Excel 97/2000/XP/2003/2007/2010 (*.xls, *.xlt, *.xla, *.xlsx, *.xlsm, *.xltx, *.xltm files)

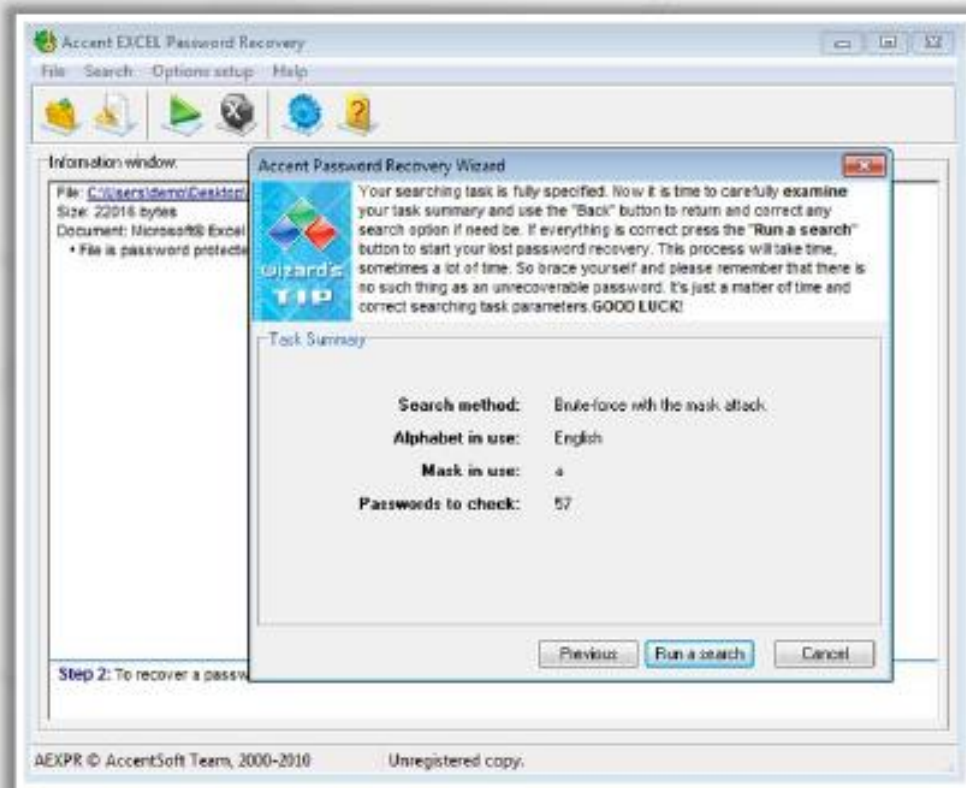
■ Features

- Removes the **"open"** MS Excel password
- Recovers or removes the **"write,"** workbook, shared workbook and worksheet passwords
- Recovers passwords of any length and complexity
- Supports **multilingual passwords**
- Protects passwords against unauthorized program access



<http://www.rixler.com>

Accent EXCEL Password Recovery



<http://www.passwordrecoverytools.com>

Accent EXCEL Password Recovery **restores lost passwords** to your Microsoft Excel documents

Features

- Supports Microsoft Excel 97, Excel 2000, Excel XP, Excel 2003, Excel 2007, and Excel 2010
- Supports different password types; password to open a document, password to modify, and spreadsheet protection password
- Recovers passwords using Brute-Force attack, mask and dictionary-based attacks



Excel Password

- Excel Password is able to recover any Excel document **regardless of the password length**
- It supports **several protection types**: password to open, password to modify, workbook protection password, worksheet protection password, etc.

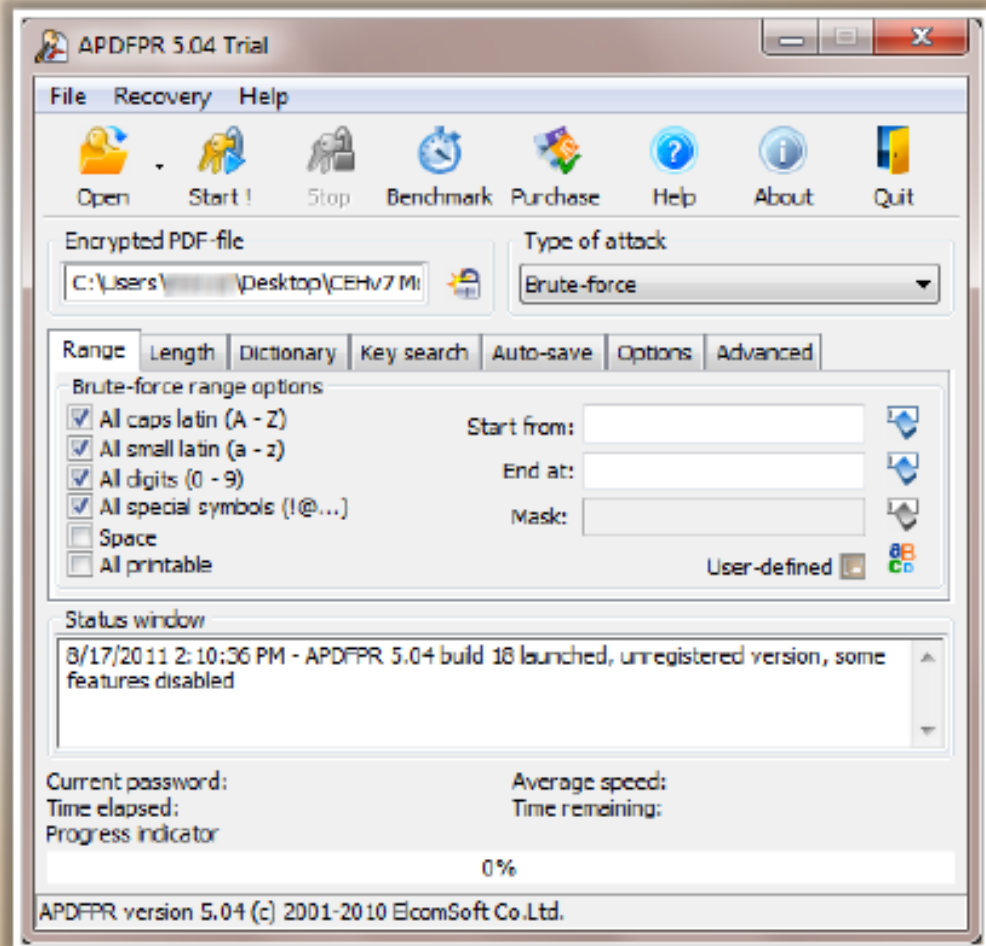


Advanced PDF Password Recovery

Advanced PDF Password Recovery **recovers** or instantly **removes** passwords protecting or locking PDF **documents** created with all versions of Adobe Acrobat or any other PDF application

Features

- Instantly **unlocks** PDF **documents** with printing, copying and editing restrictions
- Removes "**owner**" and "**user**" passwords
- Recovers** passwords to open
- Supports 40-bit and 128-bit RC4 **encryption** as well as 128-bit and 256-bit AES encryption
- Dictionary** and **Brute-Force** attacks with user-defined masks and advanced templates



<http://www.elcomsoft.com>

PDF Password Cracker

PDF password cracker is a utility to **remove** the **security** on PDF documents

Both protection methods are cracked:

- **Restricted** operations on file can be **disabled**
- It also can be used to **decrypt** files for which you know the passwords



<http://www.crackpdf.com>

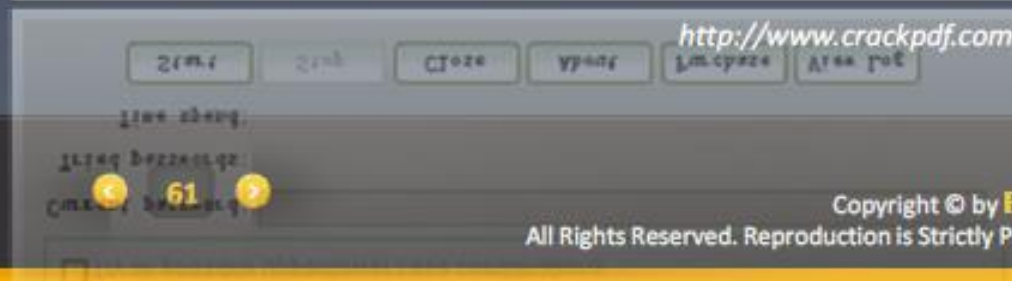
PDF Password Cracker Pro

PDF Password Cracker Pro is for **cracking PDF passwords**

It supports **Adobe** Standard 40-bit decryption, Adobe Advanced 128-bit decryption and **AES decryption**

Features

- Supports the **cracking** of both **user** and **owner** passwords
- Uses **brute-force** and **dictionary attacks**
- Saves** password **search** state when clicking pause



Atomic PDF Password Recovery



Features

- Supports files in format PDF 1.7 and earlier
- Works with PDF files encrypted using RC4 and AES algorithms
- Offers batch processing of PDF files
- Removes PDF restrictions in a fraction of a second
- Compatible with Windows systems

Atomic PDF Password Recovery **disables security setting limitations** protecting Adobe Acrobat PDF from editing, printing, selecting text and graphics, or adding/changing annotations and form fields

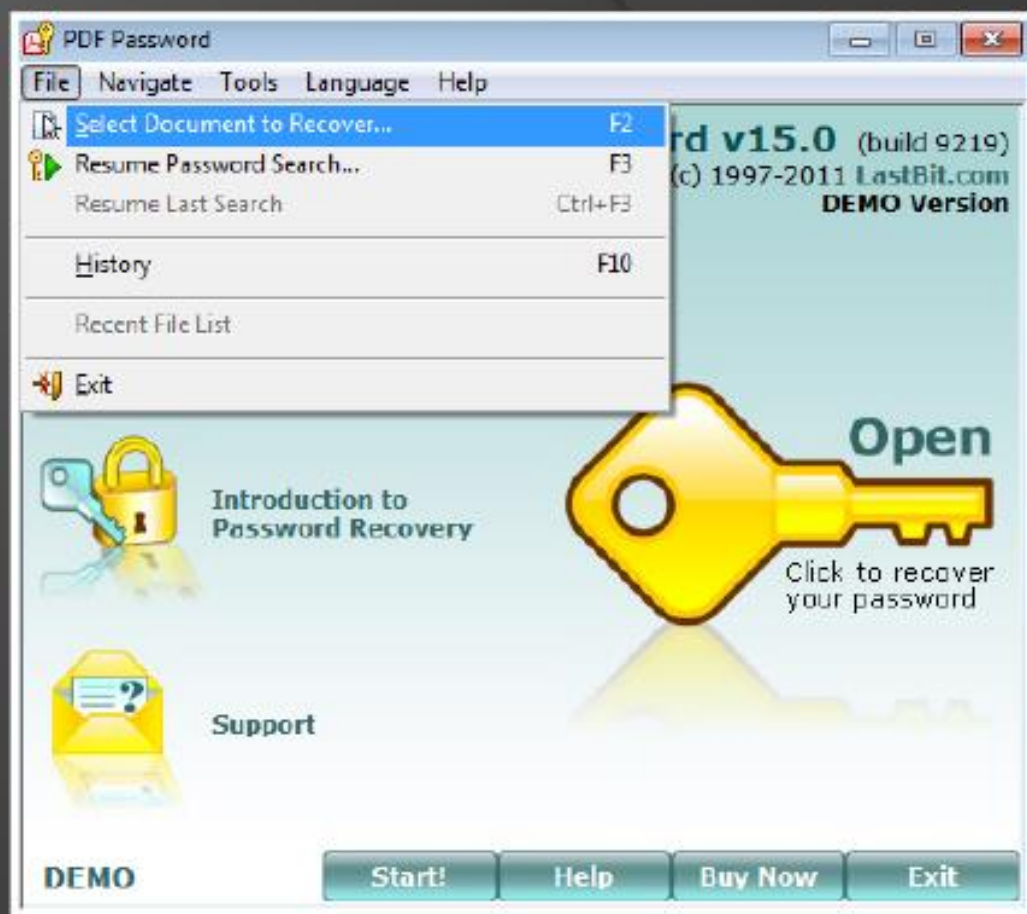


<http://apasscracker.com>

PDF Password



- PDF Password **recovers lost passwords for password-protected** PDF files
- It recovers user and permission passwords only
- PDF Password does not allow breaking **DRM** (Digital Right Management) system
- It uses various recovery methods to recover the password such as **brute-force attack, dictionary attack, smart force attack, password variation**, etc.



Recover PDF Password



Recover PDF Password **removes restrictions and recovers passwords** of protected PDF files

Features

- Open, print, copy or edit password-protected PDFs
- Get an unprotected copy of a PDF file without a password search
- Use various methods to recover PDF passwords
- Support all versions of Adobe Acrobat up to v.9



<http://www.eltima.com>

Appnimi PDF Password Recovery

Appnimi PDF Password Recovery allows users to **search for the password of the protected PDF** using a brute-force algorithm

Features

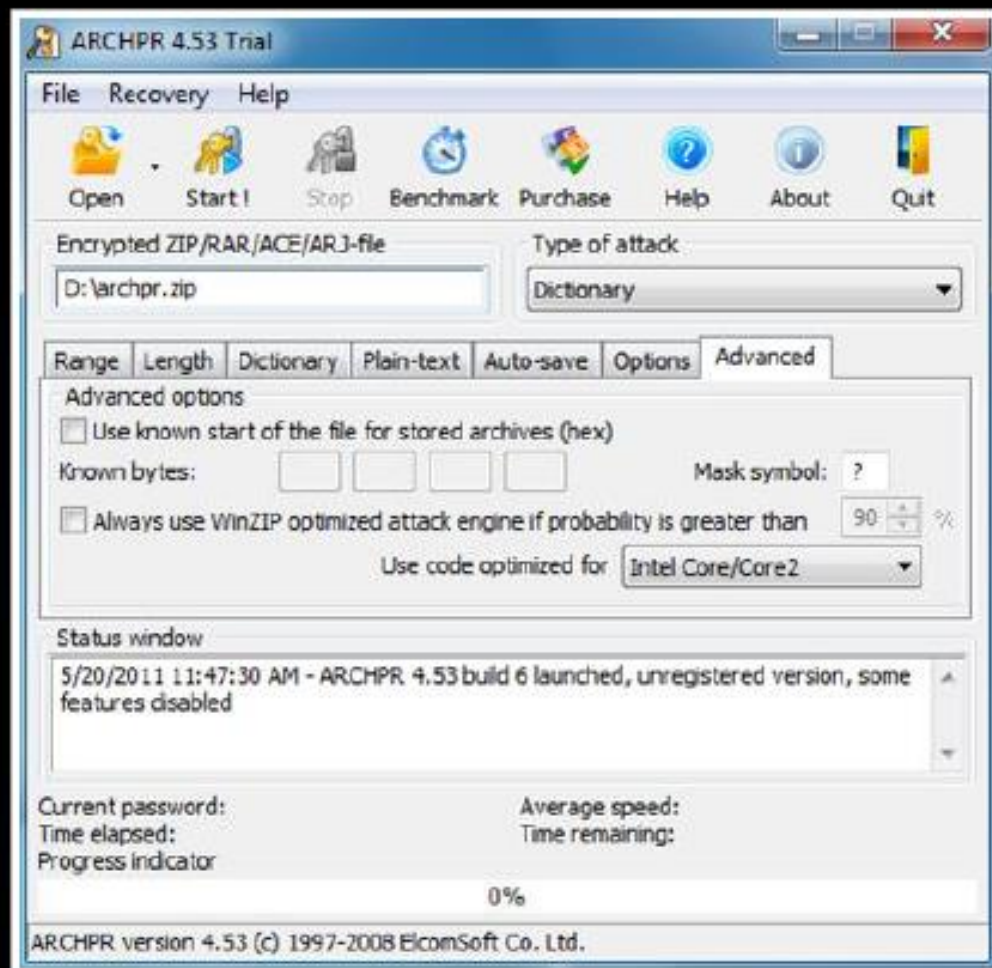
- Recovers the user password required to open the file
- Uses a brute-force algorithm to recover the password
- Allows users to set the parameters before starting the recovery process



<http://www.appnimi.com>

Advanced Archive Password Recovery

- Advanced Archive Password Recovery **recovers** protection passwords or unlocks **encrypted** ZIP and RAR archives created with all versions of popular archivers
- Recovers** passwords for **plain** and **self-extracting** archives created with PKZip and WinZip, RAR and WinRAR **automatically** or with your **assistance**



KRyLack Archive Password Recovery



KRyLack Archive Password Recovery is a program to recover lost or forgotten passwords to files of the following types:

ZIP, RAR, and ACE archives

Features

- Support all archive files created by various software packages
- Work with archives containing only one encrypted file
- Support RAR archives with encrypted filenames
- Support ZIP Archives
- Support self-extracting archives



<http://www.krylack.com>

Zip Password

- Zip Password recovers **password-protected ZIP archives**, both standalone (.zip) and self-extracted archives (.exe)
- It supports all ZIP files created by **various software packages**



Atomic ZIP Password Recovery

- Atomic ZIP Password Recovery **restores lost and forgotten passwords to ZIP archives** using classic encryption
- Classic encryption is **built in to the ZIP format** and is supported by all software for managing ZIP archives: WinZip, WinRar, PKZIP, 7-Zip, etc.



<http://apasscracker.com>



Features

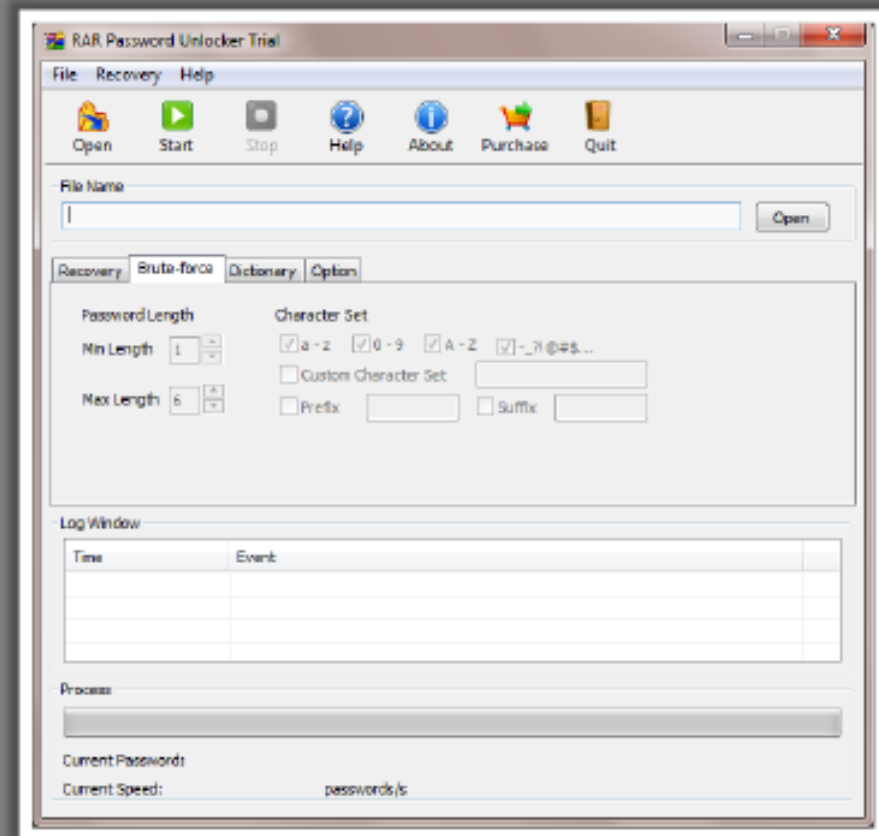
- Recovers passwords using two methods: **brute-force and dictionary attack**
- With Atomic ZIP Password Recovery, search can be paused at any point in the process and continue where it was left off at a later time
- Calculates the time required to recover the password and gathers all the information to decide on the type of attack to choose for your lost ZIP password

RAR Password Unlocker

RAR Password Unlocker tool recovers forgotten or lost password from **RAR archives encrypted by WinRAR 3.90** and its previous versions

Features

- Recovers passwords from password-protected RAR files
- Program priority is available for a higher RAR password recovery speed
- Speed up RAR password recovery with multi-core CPU
- Auto-save feature enables you to resume recovery at any time after a stop or interruption



<http://www.passwordunlocker.com>

Default Passwords

A default password is a password supplied by the manufacturer of new equipment that is password protected



Online tools that can be used to search default passwords:



1. <http://www.phenoelit-us.org>
2. <http://www.defaultpassword.com>
3. <http://cirt.net>
4. <http://default-password.info>
5. <http://www.defaultpassword.us>
6. <http://www.passwordsdatabase.com>

Vendor	Model	Version	Access Type	Username	Password
3COM	CoreBuilder	7000/6000/3500/2500	Telnet	Debug	Synnet
3COM	CoreBuilder	7000/6000/3500/2500	Telnet	Tech	Tech
3COM	HiPerARC	v4.1.x	Telnet	Adm	(none)
3COM	LANplex	2500	Telnet	Debug	Synnet
3COM	LANplex	2500	Telnet	Tech	Tech
3COM	LinkSwitch	2000/2700	Telnet	Tech	Tech
Huawei	E960			Admin	Admin
3COM	NetBuilder		SNMP		ILMI
3COM	Netbuilder		Multi	Admin	(none)
3COM	Office Connect ISDN Routers	5x0	Telnet	n/a	PASSWORD
3COM	SuperStack II Switch	2200	Telnet	debug	Synnet
3COM	SuperStack II Switch	2700	Telnet	tech	Tech
3COM	OfficeConnect 812 ADSL		Multi	adminmtd	adminmtd

<http://www.phenoelit-us.org>

http://www.defaultpassword.com

default password list

Browse by character: **A B C D E F G H I J K L M N O P Q R S T U V W X Y Z 0-9**

Search

Manufacturer: **dpl**

Product: **dpl**

Contribute to the default password list.

Add your own experience

Manufacturer: Product: Revision:

Protocol: **Multi** Access:

User ID: Password: **dpl**

Last updated 2011-05-19 11:12:22 (GMT+1)
contact at defaultpassword dot com



- wap this site, read more -

Advertising: Du kan sluta leta efter webshotell, lopnet hjälper dig..

This tool allows you to:

- **Browse** for a list of passwords in **alphabetical order**
- **Search** passwords for particular **manufacturer** or **product**
- **Provide** a **password** of your choice to be added to the **Database**

<http://www.cirt.net/passwords>

- Cirt.net **lists vendors** in alphabetical order. When a vendor in the list is clicked, a window containing a **user's information** (ID, password, product and version) is displayed.
- Also, the database can be **updated** by filling out the information and mailing it to sullo@cirt.net



http://www.defaultpassword.us

Default Password website maintains an **archive of the passwords** for various routers, software, operating systems, etc. that come with some **default factory** installed password in one way or the other



DefaultPassword.us
Oh I forgot my password

Home

Search

Main Menu

- Home
- Link Password
- Netgear Password
- Linksys Default Password
- Netcool Password
- Netopia Password
- SMC Password
- TP Link Password
- Other Password
- Zte Default Password
- Belkin Password
- Motorola Password
- Zte Password
- ZyXEL Password
- Dsl Password
- ZTE Password
- SpeedTouch Password
- SAGEM Password
- Arris Password
- Canon Default Password
- Siemens Password
- Arris Default Password

Password for

- Motorola Wimax CPE Password
- Echolife home gateway Default Password
- Netopia 3347 Password
- Link DL 624 password
- Link WBR-1310 Default Password

Default Password

The Default Password

Welcome to the Default Password website. Resetting password is simple and we do it everyday, however, some systems have default passwords, which if not changed, can be recovered easily here at this website. we maintain an archive of the passwords for various routers, software's, operating systems etc which come with some default factory installed password in one way or the other.

Please bear in mind that with this website, we are in no way promoting the use of default passwords, as they can be easily exploited by anyone. So be safe and change your passwords regularly.

Login Form

Username

Password

Remember Me ☐

Login

• Forgot your password?

• Forgot your username?

• Create an account

Reference: Netconnect

Get your own free website with SuperFast Wireless Broadband.
info@netconnect.co.uk/3004999

dfangetx

0 Link Save 40-60% on New, Surplus, Repair
www.PCPartner.com

a-Newsletter for SMS

Free access to bar tips & tricks. Subscribe now to win an iPad
www.Cheeky.com/smsnews

AdChoice

Who's Online

We have 6 guests online

Powered by Defaultpassword.us, valid XHTML and CSS.

http://www.passwordsdatabase.com

PasswordsDatabase.com

Default Passwords



391 vendors, 1686 passwords

Not Found

The requested URL /feed/ was not found on this server.
Bingo online? And every bingo sites UK, from the best bingo brand.
[Video Games Strategy](#)

[A](#)[B](#)[C](#)[D](#)[E](#)[F](#)[G](#)[H](#)[I](#)[J](#)[K](#)[L](#)[M](#)[N](#)[O](#)[P](#)[Q](#)[R](#)[S](#)[T](#)[U](#)[V](#)[W](#)[X](#)[Y](#)[Z](#)[Other](#)

Vendors

360 Systems	3COM	3M
3CC	Acc/Bridgeline	Accelerated Networks
3GCTON	Acip	actiontec
adaptec	Adaptec RAID	ADC Kambria
adCompass.com	ADP	Adtran
Advanced Integration	Adinet	Alcatel
Alcatel Thomson	Alcatel/Bridgeline/Timeslip	allied
Alfred Telecom	Alfred Telecom	alfred-telecom
Altach	Altach Web Systems	AMBT
AMT	Amigo	Amproton



Navigation

- Home
- About Us
- Terms of use
- Submit an entry

Calls, broadband and TV,
BT does all three

[Find out more](#)

BT Vision 

[AsChoices >](#) 

Teledyne TSS
Gyrocompass &
Motion Sensors
Subsea Pipe and
Cable Detection

http://www.virus.org

Virus.Org

- The Virus.org default password database was created to provide a **resource** for verified **default login/password** pairs for common networked devices
- Its password **database** contains **default** passwords for **equipment** and **software** from many vendors, including Cisco, Nortel, IBM, etc.



[Home](#)
[Links](#)
[Password Database](#)

[Main Menu](#)

- Home
- News
- Articles
- Password Database
- Waiting List Archive
- Virus.Org Designs
- Virus.Org Jabber

[New Google](#)
[SQL Calculation](#)
[Magic Rank](#)
[enSentry](#)
[World's Leading](#)
[tool for SQL](#)
[verification and](#)
[selection](#)
[www.mssql.net](#)

Virus.Org Default Password Database

The Virus.Org default password database was created to provide a resource for verified default login/password pairs for common networked devices. The goal is to document as many known cases of default login credentials on as many different devices and software packages as possible.

The logins and passwords contained in this database are either set by default when the hardware or software is first installed, or are in some cases hardcoded into the hardware or software. All logins and passwords go unchanged even when the item in question is publicly service on a public network. It is hoped that a comprehensive catalog of these logins and passwords will help ensure more accurate auditing.

The data provided is sourced from a number of sources across the Internet, including the BeautifulSoup YUL4 DEV, P04-TEST and other mailing lists. It also includes passwords gathered by our team over numerous Penetration Testing assignments.

Our database contains default passwords for equipment and software from many vendors including 3Com, Cisco, H3C, HP, Compaq, Digital, D-Link, Linksys, Oracle and Microsoft.

Vendor Product	Model/Version	Login	Password	Access Level	Comments
2wire WiFi Routers		admin	admin	Admin	Almost all 2wire routers
3COM Cellstar	7009	tech	tech		
3COM CoreRouter	7009/0000	tech	tech		
3COM CoreRouter	7009/0000	tech	tech		
3COM HPeARC	v4.1.x	admin	admin		
3COM Linksys	2580	debug	debug		
3COM Linksys	2580	tech	tech		
3COM Linksys	2580/2700	tech	tech		
3COM NetStor		admin	admin		
3COM NetStor		admin	admin		
3COM Office Connection Routers	540	tech	tech		
3COM SuperStack 5 Switch	2255	debug	debug		
3COM SuperStack 5 Switch	2160	tech	tech		
3COM		admin	admin		
3COM		admin	admin		
3COM		manager	manager		
3COM		monitor	monitor		
3COM		read	read		
3COM		secretly	secretly		
3COM		write	write		
3COM NetConnect Access Point	01.50-01	tech	tech	Admin	
3COM OfficeConnect (11T AP)		admin	admin	Admin	

Module Flow



Password
Cracking
Concepts



Types of
Password
Attacks



System
Software
Password
Cracking



Password
Cracking
Tools

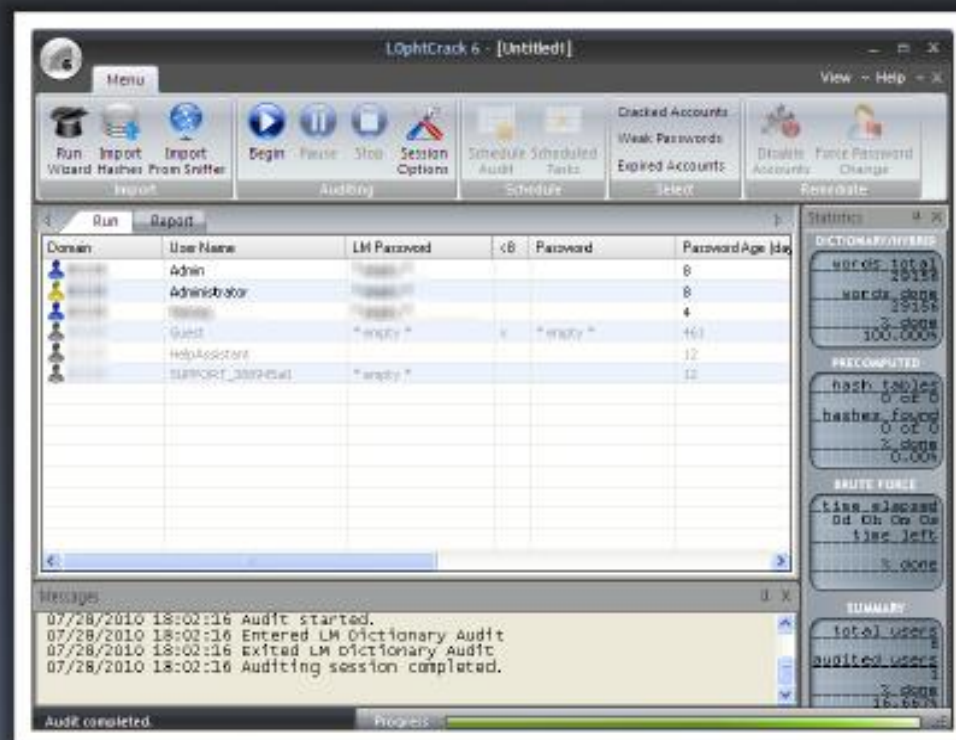


Application
Software
Password
Cracking

L0phtCrack



It has features such as **scheduling**, **hash extraction** from 64 bit Windows versions, multiprocessor algorithms, and **network monitoring** and **decoding**



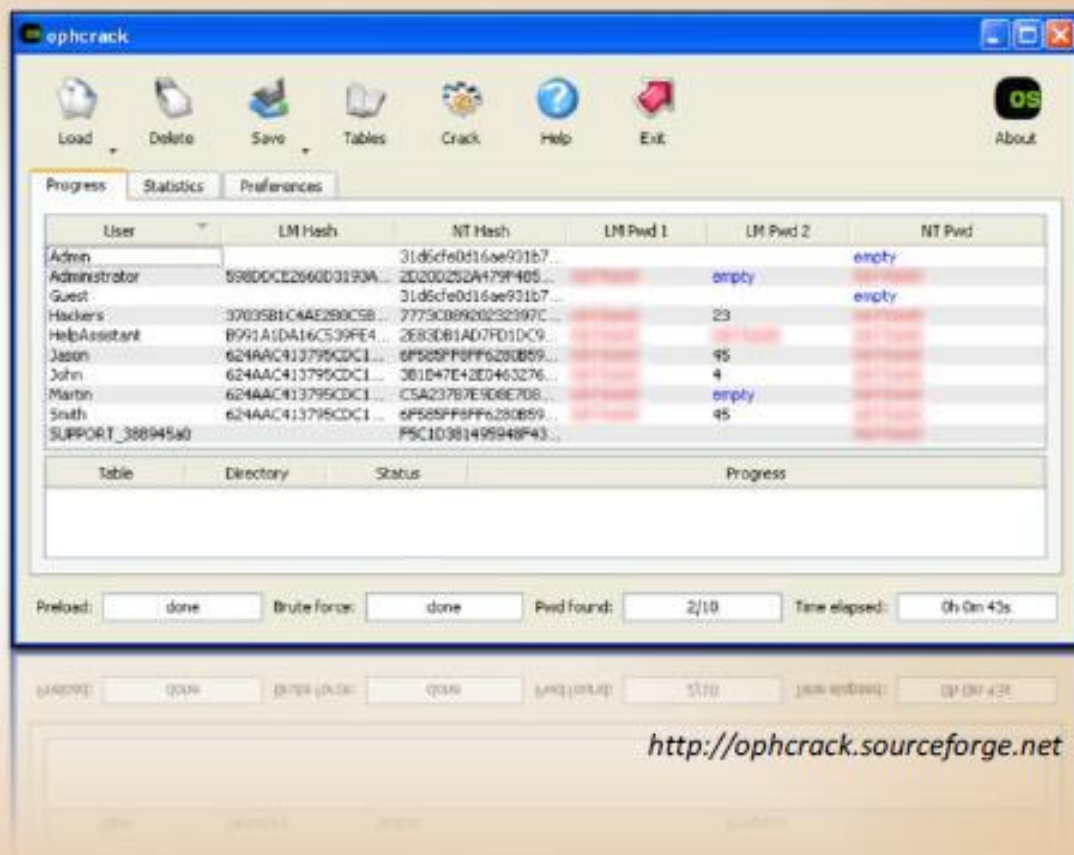
<http://www.l0phtcrack.com>

OphCrack

It is a Windows password cracker based on **rainbow tables**

Features

- Cracks **LM** and **NTLM** hashes
- Brute-force module for simple passwords
- Loads hashes from encrypted **SAM** recovered from a Windows partition



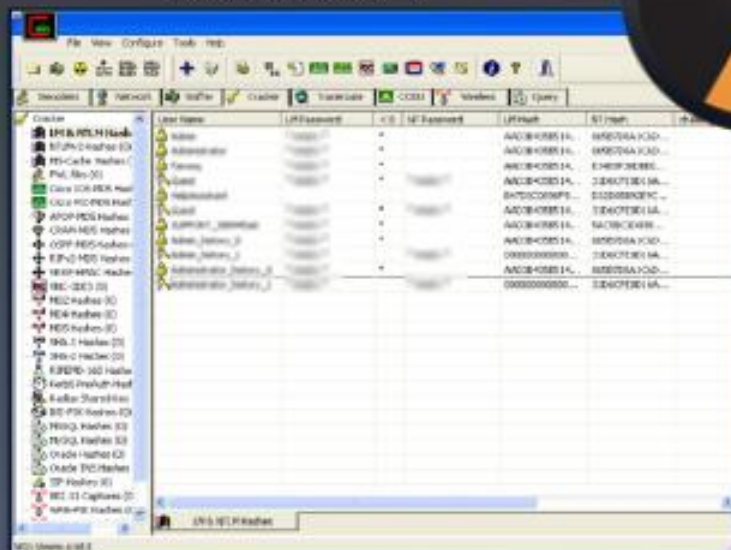
<http://ophcrack.sourceforge.net>

Cain & Abel



Cain & Abel is a **password recovery tool** for Microsoft Operating Systems

It allows **recovery of various kind of passwords** by sniffing the network, cracking encrypted passwords using dictionary, brute-force and cryptanalysis attacks, recording VoIP conversations, decoding scrambled passwords, recovering wireless network keys, revealing password boxes, uncovering cached passwords and analyzing routing protocols

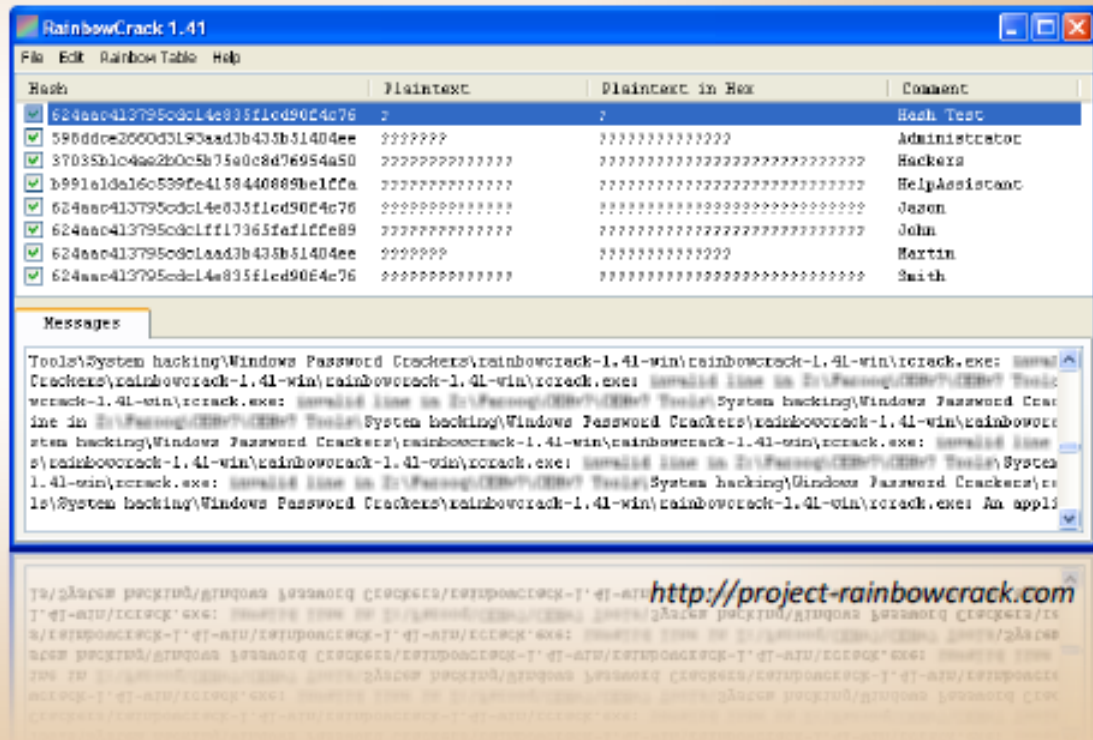


<http://www.oxid.it>

RainbowCrack



RainbowCrack uses a **time-memory tradeoff algorithm** to crack hashes. It differs from the hash crackers that use a brute-force algorithm.



Windows Password Unlocker

■ Password Unlocker resets **unknown or lost Windows local administrator, domain administrator** and other user account passwords on all popular Windows OS

■ It allows you to easily get instant access to your locked computer without **reinstalling Windows** and resorting to expensive PC engineers



<http://www.passwordunlocker.com>

Windows Password Breaker



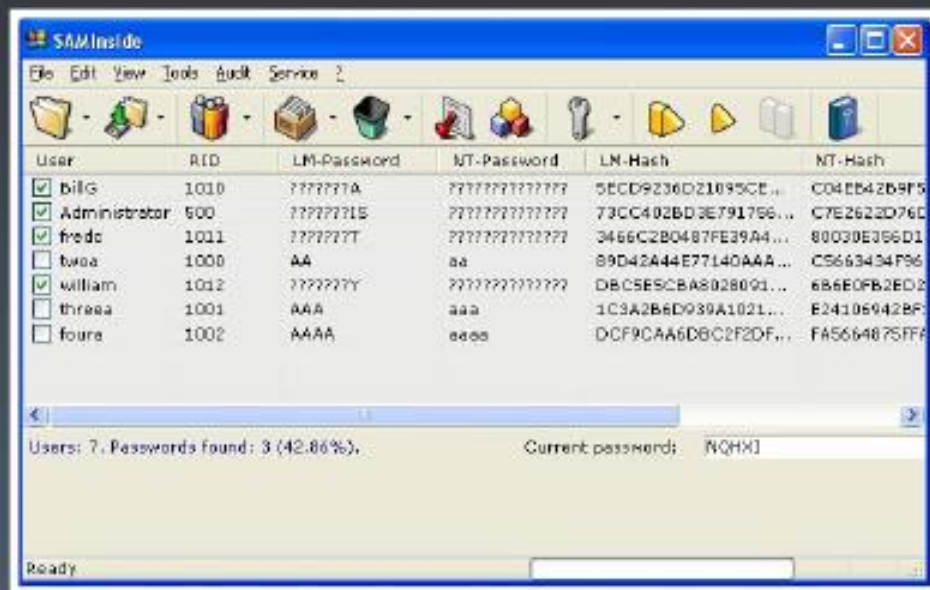
- Windows Password Breaker **resets or recovers any lost, forgotten or unknown** Windows local administrator, standard or guest users, and Windows domain administrator passwords
- It automatically burns a password recovery disk with a **CD/DVD or USB flash drive**



<http://www.recoverwindowpassword.com>

SAMInside

SAMInside **recovers the user passwords of**
all Windows operating systems



It can be run from a
diskette, CD/DVD-disk
or **USB-drive**



It uses brute-force attack, distributed attack,
mask attack, dictionary attack, hybrid attack,
and, pre-calculated tables attack to **recover the**
password



PWdump7 and Fgdump



Attacker

pwdump7.exe



```
fgdump.exe -h 192.168.0.10  
-u AnAdministrativeUser -p  
14mep4ssw0rd
```

Dumps a remote machine (192.168.0.10)
using a specified user

Fgdump works like
pwdump but also
extracts cached
credentials and
allows remote
network execution

PWDUMP extracts LM
and NTLM password
hashes of local user
accounts from the
Security Account
Manager (SAM)
database

This tool must be
run under an
administrator
account

Password Cracking Tools



PCLoginNow

<http://www.pclloginnow.com>



Proactive System Password Recovery

<http://www.elcomsoft.com>



KerbCrack

<http://ntsecurity.nu>



Password Unlocker Bundle

<http://www.passwordunlocker.com>



Recover Keys

<http://recover-keys.com>



Windows Password Reset Professional

<http://www.resetwindowspassword.com>



Windows Password Cracker

<http://www.windows-password-cracker.com>



Windows Password Reset Standard

<http://www.resetwindowspassword.com>

Password Cracking Tools



krbpwguess

<http://www.cqure.net>



Rockxp

<http://www.korben.info>



Password Kit

<http://www.windowpassword-recovery.com>



PasswordsPro

<http://www.insidepro.com>



WinPassword

<http://lastbit.com>



LSASecretsView

<http://www.nirsoft.net>



Passware Kit Enterprise

<http://www.lostpassword.com>



LCP

<http://www.lcpsoft.com>

Password Cracking Tools



MessenPass

<http://www.nirsoft.net>



Protected Storage PassView

<http://www.nirsoft.net>



Mail PassView

<http://www.nirsoft.net>



Network Password Recovery

<http://www.nirsoft.net>



Messenger Key

<http://www.lostpassword.com>



Asterisk Key

<http://www.lostpassword.com>



Dialupass

<http://www.nirsoft.net>



IE PassView

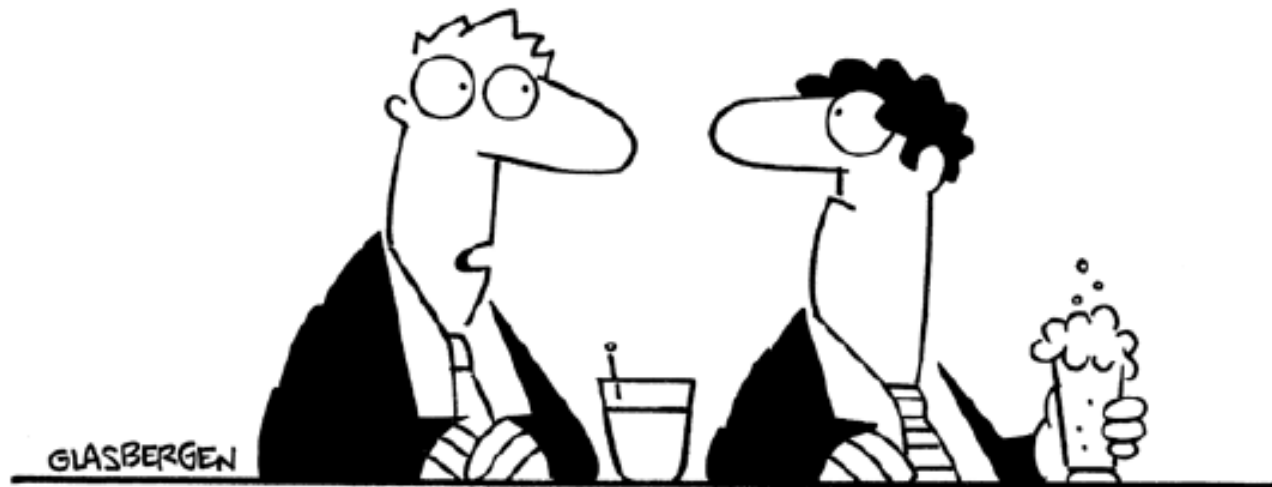
<http://www.nirsoft.net>

Module Summary



- ☐ A password is a secret series of characters created to secure important files from unauthorized access
- ☐ Password cracker is any program that can decrypt passwords or otherwise disable password protection
- ☐ Cracking is a way to escalate security of the program or a system
- ☐ Crackers use various password cracking methods such as brute-force, dictionary attack, syllable attack, rule-based attack, distributed network attack, and guessing
- ☐ Password cracking software are classified into system software password cracking and application software password cracking
- ☐ Run a cracker periodically on your own password files to check the password strength

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www.glasbergen.com



“While I was thinking outside of the box, someone changed the password and now I can’t get back in!”

Copyright 2004 by Randy Glasbergen.
www.glasbergen.com



**"I created a password and wrote it down like you told me to.
Then I locked it away in a secure folder for safekeeping.
But I need my password to get into the folder!"**